

Estimating SME Banking Wallet Size and Product Propensity: A Survey of Relevant Literature and Modeling Approaches

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Abstract

Retail banks that serve small and medium-sized enterprises (SMEs) often observe only a partial view of a business customer’s financial life. A bank may know its own deposits, card spend, loans, treasury activity, and relationship history, but it rarely observes the customer’s balances and product holdings at competing institutions. This creates a practical “wallet size” problem: estimate the products and balances an SME holds with competitors, with deposits as the main business object, and estimate how much of that wallet can realistically be moved to the focal bank. This survey reviews literatures that bear on the problem: cross-selling and next-product-to-buy models in financial services, share-of-wallet and customer-share models, structural bank deposit demand and bank competition, SME relationship lending, and causal response/uplift modeling. The main conclusion is that the exact end-to-end problem is not a single canonical academic model. It is better viewed as a composite decision system that combines latent wallet reconstruction, deposit and product demand estimation, offer-response modeling, and constrained profit optimization.

1 Introduction

For a US retail bank serving SMEs, the commercially important question is deposit-led wallet opportunity. The bank wants to estimate the total financial wallet of the business, how much of that wallet sits with competitors, and how much of the competitor-held wallet is realistically movable to the focal bank. Deposits are the anchor product because they provide funding, transactional information, and a relationship platform for payments, lending, and treasury services. Other products matter, but the main use case in this paper is deposit-led wallet sizing rather than product-assortment design.

The literature does not usually call this problem “wallet size estimation.” Relevant work appears under several labels: cross-selling, cross-buying, next-product-to-buy, customer share, share-of-wallet, bank deposit demand, bank competition, and relationship lending. The practical modeling problem is therefore interdisciplinary. Marketing science contributes models of customer product ownership and campaign response. Industrial organization and banking economics contribute models of bank choice, deposit demand, and competition. SME finance contributes theories of relationship lending and the informational role of transaction accounts. Causal inference contributes methods for separating who is likely to buy from who is likely to buy because of a marketing action.

This paper surveys these related literatures, organizes the model architectures that they imply, and then recommends an integrated architecture for a bank that wants to estimate competitor deposit wallet size and capture potential.

2 The SME Wallet-Size Problem

Let i index an SME firm, b index a bank, and p index products such as demand deposits, savings deposits, credit cards, commercial real estate loans, term loans, lines of credit, merchant services, payroll, treasury management, and investment funds. Let

$$y_{ibp} \in \{0, 1\}$$

denote whether firm i holds product p at bank b , and let

$$a_{ibp} \geq 0$$

denote the associated size, such as average deposit balance, outstanding loan balance, card spend, or assets under management. The total product-specific wallet is

$$W_{ip} = \sum_b a_{ibp}.$$

For the focal bank f , the observed part is a_{ifp} and y_{ifp} . The competitor component

$$W_{ip}^{-f} = \sum_{b \neq f} a_{ibp}$$

is generally missing or only indirectly observed.

In deployment, these objects should carry a time index because both product need and product size move over time. Write

$$A_{ipt}^{-f} = \sum_{b \neq f} a_{ibpt}$$

for the competitor-held amount of product p at decision time t . The business question is not only whether A_{ipt}^{-f} is large. It is how much of that amount is addressable by a specific offer. Define

$$A_{ipo}^{\text{addr}}(H) = M_{ipo}(H) A_{ipt}^{-f}, \quad 0 \leq M_{ipo}(H) \leq 1,$$

where $M_{ipo}(H)$ is the structural offer- and horizon-specific movable fraction. This is not a generic purchase propensity. It is the fraction of competitor-held balances, borrowing, spend, or assets that can plausibly migrate to the focal bank under offer o over horizon H . Operational reports may also show ratios such as moved dollars divided by estimated competitor-held dollars; those reporting ratios should be distinguished from the latent structural fraction $M_{ipo}(H)$.

The decision problem has three parts. First, the bank wants an estimate of latent wallet state:

$$\Pr(y_{ibp} = 1 \mid X_i, Z_{bt}, A_{bpt}, M_t), \quad \mathbb{E}[a_{ibp} \mid X_i, Z_{bt}, A_{bpt}, M_t, y_{ibp} = 1],$$

especially for competitor banks $b \neq f$, where Z_{bt} and A_{bpt} summarize bank and bank-product attributes observed or imputed by the model. Second, it wants the distribution of product amounts already held at competitors, A_{ipt}^{-f} , and the addressable amount $A_{ipo}^{\text{addr}}(H)$. Third, for an offer or marketing action o —for example a deposit-rate offer, relationship-manager outreach, onboarding support, or operating-account proposal—the bank wants outcome-specific incremental effects. For a generic outcome Y ,

$$\tau_{ipo}^Y = \mathbb{E}[Y_{ip}(o) - Y_{ip}(0) \mid \mathcal{I}_{it}],$$

where Y may be product ownership, balance dollars, spend, risk-adjusted relationship value, runoff, or retention. Balance lift, moved competitor dollars, and value lift are therefore related but distinct estimands.

The distinction between prediction and causal response is crucial. A model can predict that a firm is likely to buy a loan or maintain large deposits without showing that a campaign will cause the firm to move wallet to the focal bank. It can also estimate a large competitor-held wallet without identifying how much of that wallet is movable. The literature reviewed below supports different parts of this system.

3 Literature Survey

This section groups the main bodies of literature relevant to SME wallet estimation. The key point is that the literature does not support a single static “wallet size” object. It supports several linked objects: latent product need, observed product ownership, customer share, bank choice, dynamic liquidity demand, and causal marketing response.

3.1 Cross-Selling and Product-Ownership Models

The closest direct literature is financial-services cross-selling. Early work by Kamakura et al. [37] applies latent trait analysis to evaluate prospects for cross-selling financial services. This line of work treats observed product holdings as noisy signals of latent customer needs, financial maturity, or product propensity. For a bank, this is directly relevant because competitor holdings are often latent and must be inferred from partial signals.

Kamakura et al. [38] extend the database-marketing perspective using a mixed-data factor-analyzer approach for data augmentation and prediction. The important practical insight is that product holdings, demographics, and monetary variables can be modeled jointly, so that missing or unobserved product positions are not handled as independent binary classification problems. This matters for SME wallet estimation because deposits, lending, merchant services, and cards are correlated through firm size, cash-flow volatility, owner wealth, industry, life-cycle stage, and bank relationship depth.

Another directly relevant stream is next-product-to-buy (NPTB) modeling. Knott et al. [41] describe NPTB models for cross-selling applications and evaluate their use in a field setting. In banking, an NPTB model can rank likely products for each SME relationship. The limitation is that a next-product model usually optimizes product targeting, not total competitor wallet reconstruction. It answers “what is the next product to offer?” more directly than “how large is the competitor-held deposit wallet?”

Li et al. [44] study cross-selling sequentially ordered products in consumer banking services using a structural multivariate probit framework. The paper is especially useful because banking products may follow life-cycle or business-maturity sequences: transaction accounts precede cards and merchant services; operating deposits may precede credit lines; treasury services may follow scale and complexity. For SME banking, the sequence is not identical to household financial maturity, but the modeling principle carries over: product ownership is dynamic and correlated across products.

3.2 Share-of-Wallet and Customer-Share Models

The share-of-wallet literature is relevant when the focal bank already has a relationship with the SME and wants to estimate the fraction of total financial activity captured by the bank. Verhoef

[65] examines how customer relationship perceptions and marketing instruments affect retention and customer share development over time. This literature separates retention from depth: keeping a customer is not the same as gaining a larger share of that customer’s wallet.

Keiningham et al. [40] study customer satisfaction and actual share-of-wallet in a business-to-business environment. This is relevant because SME banking is closer to relationship-managed B2B service than to anonymous retail product purchase. Satisfaction, relationship quality, and switching costs can affect whether an SME consolidates deposits, lending, cards, and treasury activity at one bank.

For the bank’s use case, customer-share models can be written as

$$s_{ipf} = \frac{a_{ifp}}{W_{ip}},$$

where s_{ipf} is the focal bank’s share of the SME’s product- p wallet. When W_{ip} is unknown, the model must combine internal balances with external signals, market benchmarks, credit bureau information, payments data, public records, firmographics, and statistical priors. This makes the share-of-wallet problem partly a missing-data problem and partly a market-demand problem.

3.3 Bank Deposit Demand and Competition

Deposit wallet estimation cannot be reduced to generic cross-selling because deposits have special economic features. Deposits are funding sources for banks, liquidity services for customers, and relationship anchors. They also respond to rates, fees, branch and digital convenience, safety perceptions, payment needs, and switching frictions.

Dick [23] estimates demand and welfare in the banking industry, providing a structural demand perspective for bank services. Adams et al. [3] study competition among depository institutions, which is important for defining the relevant competitive set. For an SME in a local market, the competitors may not be all banks in the country; they may be nearby branch banks, digital banks, specialized lenders, card issuers, fintech payment providers, and industry specialists.

Egan et al. [26] estimate deposit competition and financial fragility in the US banking sector. Their work is relevant because deposit demand and bank competition affect how much of a customer’s balances can realistically be moved by price or relationship offers. Drechsler et al. [24] study the deposits channel of monetary policy, emphasizing the role of deposit market power and rate pass-through. For a wallet-size system, these papers motivate models that include deposit rates, local competition, account type, market concentration, and customer rate sensitivity.

A practical deposit model often needs a two-part structure:

$$\Pr(a_{if,\text{dep}} > 0 \mid X_i)$$

for deposit relationship incidence and

$$\mathbb{E}[\log a_{if,\text{dep}} \mid a_{if,\text{dep}} > 0, X_i]$$

for conditional deposit size. If competitor balances are unobserved, the model may use market-level deposit data, firm revenue estimates, industry working capital ratios, payments volume, and observed inflows/outflows as indirect signals of total deposit wallet.

3.4 Dynamic Deposit Demand

A major weakness of the wallet-size analogy is the implicit constant-size assumption. For deposits, the assumption is particularly fragile. An SME’s deposit balance is not only a product holding; it

is a time-varying operating balance. It moves with revenues, payroll, vendor payments, seasonality, business growth, and the normal timing of cash inflows and outflows. A firm can need the same banking product in two periods but require very different deposit balances.

The corporate liquidity literature gives a useful framework. Opler et al. [53] study determinants of corporate cash holdings and emphasize transaction and precautionary motives. Almeida et al. [4] develop the idea that financially constrained firms save cash out of cash flows, while Han and Qiu [31] study precautionary cash holdings under cash-flow risk. Bates et al. [7] show that cash holdings of US firms increased over time and relate this to riskier cash flows and changing firm characteristics. These papers are mostly about corporate cash rather than bank deposit accounts, but for SME banking the connection is useful: much of an SME’s operating liquidity is expressed as bank deposits or near-deposit balances.

Private-firm and smaller-firm evidence is especially relevant. Mortal et al. [49] compare cash holdings of private and public firms internationally. Poti et al. [56] and Lefebvre [42] study precautionary cash motives in private and smaller firms. These papers support the view that a deposit target should vary across industries, firm sizes, geographies, and business cycles rather than being treated as a fixed customer attribute. The more specialized question of deposit pricing and stress-period behavior is discussed separately below.

This literature suggests that the focal object should be dynamic deposit demand, not fixed deposit wallet size. A more appropriate target is

$$D_{it}^* = g(X_{it}, M_t, R_{it}, C_{it}, B_{it}, \eta_i) + \varepsilon_{it},$$

where D_{it}^* is desired liquidity or deposit balance, X_{it} are firm-level cash-flow and product variables, M_t are market conditions, R_{it} captures rates and fees, C_{it} captures financing conditions, B_{it} captures bank relationship and switching frictions, and η_i is firm heterogeneity. Observed deposits are then a partial allocation of this desired liquidity across banks and instruments:

$$D_{ift} = s_{ift} D_{it}^*,$$

where s_{ift} is the focal bank’s dynamic share of deposit liquidity. In this view, wallet estimation decomposes into two moving parts: the time-varying size of the firm’s liquidity demand and the focal bank’s share of that demand.

3.5 SME Relationship Lending and the Role of Transaction Accounts

SME banking differs from mass retail banking because information asymmetry is central. Relationship lending theory explains why small firms benefit from long-lived bank relationships and why transaction data can be valuable for credit decisions. Petersen and Rajan [54] and Berger and Udell [8] are foundational references on small-business lending relationships and lines of credit.

Transaction accounts are particularly important. Mester et al. [48] study transaction accounts and loan monitoring. Norden and Weber [52] examine credit-line usage, checking-account activity, and default risk. These papers motivate a wallet model in which deposits and payments are not merely products to be sold; they are information channels that can improve underwriting, monitoring, and relationship profitability.

For an SME bank, this means a deposit offer may have value beyond spread income. Moving an SME’s operating account to the focal bank can reveal cash-flow seasonality, payroll cycles, customer concentration, merchant activity, and early distress signals. A product-propensity model should therefore be joined with a relationship-value model.

3.6 Causal Response and Uplift Modeling

Cross-sell and wallet models estimate what an SME appears likely to buy or hold. Marketing decisions require a causal estimand: what the SME will buy because of the offer. A high-propensity customer may buy without being contacted; a low-propensity customer may be expensive to convert; and an offer may shift balances without increasing total relationship value. The relevant target is incremental contribution, not raw propensity:

$$\tau_{ip}(x) = \mathbb{E}[Y_{ip}(1) - Y_{ip}(0) \mid X_i = x],$$

where Y_{ip} may be product adoption, deposit balance, risk-adjusted revenue, or lifetime value.

The modern causal-ML literature provides tools for heterogeneous treatment effects. Radcliffe and Surry [57] and Lo [45] discuss uplift and true-lift modeling in marketing contexts. Athey and Imbens [6] introduce recursive partitioning for heterogeneous causal effects, and Wager and Athey [66] develop causal forests. These methods are useful when a bank has randomized campaigns, quasi-random offer variation, or credible experiments. Without such variation, a response model risks confusing selection with treatment effect.

3.7 Model Architectures Implied by the Literature

The preceding literatures imply several architectures. These architectures are not mutually exclusive. They answer different parts of the SME wallet problem, and the main design question is how to combine them without confusing latent need, observed product ownership, competitive share, and causal response.

Architecture 1: Latent Wallet Reconstruction

The first architecture treats competitor wallet as a latent variable. A hierarchical model can describe product incidence and size jointly:

$$\begin{aligned} z_i &\sim p(z_i \mid X_i), & y_{ibp} &\sim \text{Bernoulli}(\pi_{bp}(z_i, X_i)), \\ \log a_{ibp} \mid y_{ibp} = 1 &\sim \mathcal{N}(\mu_{bp}(z_i, X_i), \sigma_{bp}^2), \end{aligned}$$

where z_i represents latent firm scale, financial sophistication, liquidity need, credit demand, and relationship preference. This architecture is closest in spirit to latent trait and mixed-data factor models in the cross-selling literature [37, 38]. It also follows the econometric logic of two-part or hurdle models: product incidence and positive product size are distinct stochastic objects, so a bank should not convert a product-propensity score into dollars without a conditional amount model [20].

For implementation, a bank could start with a simpler generalized linear mixed model or gradient-boosted two-part model and then move toward Bayesian hierarchical or variational models as the data maturity improves. The output is a posterior distribution for total wallet, competitor-held wallet, and product amount conditional on ownership, not just a point estimate. If positive amounts are modeled on the log scale, retransformation to dollar levels should be validated rather than handled by naive exponentiation; smearing-style corrections are a standard remedy [25].

Architecture 2: Product Graph and Next-Best-Action System

The second architecture treats products as a graph or sequence. Product holdings form a vector

$$\mathbf{y}_i = (y_{i1}, \dots, y_{iP}),$$

and the model predicts the next product or bundle conditional on the current state. This can be implemented using multivariate probit, discrete hazard models, recurrent models, or gradient-boosted ranking models. The academic foundation comes from cross-selling and NPTB models [41, 44].

For SME banking, the graph should distinguish operating products from credit products and owner/household products. For example, a business checking account may lead to merchant services and payroll, while a line of credit may lead to equipment finance or treasury management. A product graph is useful for sales prioritization but insufficient for deposit sizing unless paired with balance models. For the current project, this architecture is secondary; it is included as background because it explains product correlations, not because assortment or bundle choice is the main objective.

Architecture 3: Structural Deposit Demand and Competitive Response

The third architecture models the SME’s bank choice and deposit allocation as a competitive demand problem. A simplified random-utility model is

$$U_{ibp} = \alpha_i r_{bp} - \phi_i F_{bp} + \gamma_i Q_{bp} - \kappa_i C_{ib} + \epsilon_{ibp},$$

where r_{bp} is the rate or yield, F_{bp} fees, Q_{bp} service quality, and C_{ib} switching or relationship cost. Structural banking papers motivate including local competition, bank characteristics, and market power [3, 23, 24, 26].

This architecture is most useful for pricing and deposit offers. It can answer questions such as: how much incremental deposit balance should be expected if the bank offers a rate premium, fee waiver, or improved onboarding support? Its weakness is data intensity: credible estimation often needs variation in rates, fees, markets, and competitor conditions.

Architecture 4: Dynamic Panel and Time-Series Deposit Demand

Because deposits are balances observed over time, a bank should not rely only on cross-sectional wallet models. A dynamic panel model can separate firm-specific liquidity level, time-varying business conditions, market-wide stress, and relationship share. A practical specification is

$$\log D_{ift} = \rho \log D_{if,t-1} + \beta' X_{it} + \theta' M_t + \psi' O_{ift} + \alpha_i + \delta_t + u_{ift},$$

where D_{ift} is the focal-bank deposit balance, X_{it} includes observed firm cash-flow and transaction features, M_t includes macro and market conditions, O_{ift} includes the bank’s offer and relationship terms, α_i captures persistent firm heterogeneity, and δ_t captures common shocks. This model can be extended with industry-by-time effects, geography-by-time effects, state-space components, or regime-switching terms for stress periods.

If the target is total deposit demand rather than focal-bank deposits, the bank can model

$$\log D_{it}^* = \rho \log D_{i,t-1}^* + \beta' X_{it} + \theta' M_t + \alpha_i + \delta_t + \nu_{it}$$

as a latent state and then model the focal bank’s observed share:

$$\text{logit}(s_{ift}) = \gamma' Z_{ift} + \mu_i + \xi_t.$$

This separates the question “how much liquidity does the SME want now?” from “what share of that liquidity sits with our bank?” It also allows stress periods to raise total precautionary balances while the focal bank’s share may rise, fall, or stay flat.

Time-series methods are useful, but they should usually be panel time-series methods rather than a separate univariate model for each SME. Individual SME series can be short, seasonal, lumpy, and affected by idiosyncratic events. A hierarchical panel, state-space, or mixed-effects model can pool information across similar firms while allowing customer-specific dynamics. For forecasting, machine-learning sequence models can be useful, but they should be benchmarked against transparent baselines such as seasonal naive forecasts, autoregressive distributed-lag models, and fixed-effects panel regressions.

Architecture 5: Causal Uplift and Incremental Value

The fifth architecture starts from experiments or quasi-experiments. The bank randomizes or otherwise credibly varies offers and estimates heterogeneous treatment effects. The targeting rule is then based on expected incremental profit:

$$\text{NPV}_{it} = \sum_p m_p \hat{\Delta}_{ipt} - c_t - \lambda \widehat{\text{Risk}}_{it},$$

where m_p is product margin or relationship value, c_t is campaign and incentive cost, and the risk term captures credit, liquidity, operational, and conduct constraints. Uplift and causal-forest methods are natural candidates [6, 45, 57, 66].

This architecture is the most defensible for deciding whom to contact and what to offer, because it targets incremental response rather than raw propensity. Its prerequisite is disciplined experimentation and holdout design.

Architecture 6: Integrated Decision System

The most complete architecture combines the preceding pieces:

1. Estimate total deposit wallet and competitor-held balances with a latent wallet model, and treat deposit wallet as time varying.
2. Use product-adoption and product-graph models only as supporting signals for firm scale and relationship context.
3. Estimate dynamic deposit demand using panel time-series models that include firm cash-flow, seasonality, macro conditions, rates, and local market conditions.
4. Estimate deposit-specific response to rates, fees, service, and relationship actions with a demand model.
5. Estimate the movable fraction of competitor-held balances, borrowing, spend, or assets from conversion events, account-transfer evidence, relationship migrations, and experiments.
6. Estimate causal incremental lift from campaigns using randomized holdouts.
7. Optimize offers under profitability, credit-risk, liquidity, relationship-manager capacity, fairness, and compliance constraints.

This system should produce uncertainty intervals and decision diagnostics, not only point scores. Relationship managers need to know whether a prospect has a high expected deposit wallet, a high probability of product need, high incremental responsiveness, or merely high observed similarity to existing large customers.

4 Data Sources and Feature Families

The data problem is at least as important as the model class. No single public or commercial data set observes the full SME wallet at competitor banks. The bank therefore needs a layered data design: internal customer data identify the focal-bank relationship, public data anchor local markets and stress conditions, third-party and consented data provide partial external-wallet signals, and experiments or conversion events identify move-over and incrementality. The Federal Reserve data portal is useful for this program because it provides official macro-financial, banking, interest-rate, lending, and payments data [12]. Its role, however, is mostly to supply market conditions, rate regimes, credit conditions, payment trends, and stress context. It does not directly identify an individual SME’s deposits at competitor banks.

4.1 Required Data Roles

The literature implies that an SME wallet-size model should combine several types of data, each with a different identification role:

1. **Internal relationship data.** Observed deposits, loan balances, card spend, merchant transactions, treasury services, tenure, service events, relationship-manager notes, customer-service history, onboarding channel, historical offers, quoted rates, booked rates, fee waivers, and exceptions. These data define the focal-bank labels and the information set available at decision time.
2. **Transaction and operating-flow signals.** ACH, wire, payroll, merchant, check, card, lock-box, remote-deposit, and cash-flow patterns can proxy firm scale, seasonality, payroll intensity, supplier/customer networks, and external-bank relationships. Inflow and outflow volatility, payroll coverage, tax-payment timing, merchant settlement volume, and counterparty concentration are especially important for estimating total liquidity demand.
3. **Firmographics and local business structure.** Industry, geography, firm age, revenue band, payroll size, employee count, legal form, ownership, seasonality, and local establishment density. Public sources such as County Business Patterns, Business Dynamics Statistics, Business Formation Statistics, and QCEW help calibrate industry-by-geography opportunity and business-cycle exposure [61–64].
4. **Credit and public records.** Liens, UCC filings, commercial real estate loans, commercial credit bureau data, SBA lending records, property records, bankruptcy events, and public registrations where legally and contractually usable. These data are useful for product incidence and size, especially loans, lines, collateralized credit, commercial real estate exposure, and distress events.
5. **Market, competitor, and branch data.** Local branch density, branch-level deposit market share, competitor footprint, advertised rates, fee schedules, product availability, digital-bank and fintech presence, and local competitive intensity. FDIC Summary of Deposits is a natural public anchor for branch-level deposits [28]. The limitation is that such aggregates usually mix consumer, small-business, commercial, and wholesale deposits, so they should be used as noisy market-level anchors rather than direct SME wallet labels.
6. **Rate and stress context.** Federal Reserve H.8 gives aggregate commercial-bank balance-sheet conditions, H.15 gives selected interest rates, and the Senior Loan Officer Opinion Survey gives

credit-standard and loan-demand context [14–16]. Call Reports from FDIC or FFIEC provide bank-level balance-sheet and funding information [27, 29]. These data support market regimes, competitor funding pressure, stress scenarios, and deposit-pricing variables.

7. **Payments and commerce benchmarks.** Federal Reserve payments data, card-network benchmarks, merchant-acquirer data, payroll-provider signals, and industry payment benchmarks help translate operating activity into expected deposit balances and card or merchant-wallet opportunity [13]. They are usually aggregate or vendor-provided, so they need reconciliation against internal transaction data.
8. **Validation anchors.** Consented account aggregation, customer-provided statements, treasury onboarding documents, balance-transfer records, payoff letters, direct-deposit or ACH rerouting, account-closure evidence, M&A onboarding files, and post-conversion consolidated balances are the most valuable data for validating competitor wallet estimates. Without these anchors, the model may rank prospects but should not claim precise competitor-dollar estimates.
9. **Campaign, quote, and experiment data.** Offer eligibility, contact channel, price, incentive, quoted and unquoted terms, relationship-manager activity, randomized holdouts, quasi-random rate cells, and rejected offers. These data identify price sensitivity, causal uplift, move-over, cannibalization, and runoff.

4.2 How Specific Public Data Help

Table 1 summarizes the most useful public and semi-public sources. The important point is not that these data replace bank data. They provide anchors, controls, market states, and validation benchmarks that make the internal model less circular.

4.3 Data Architecture for the Recommended Model

For the recommended architecture, data should be organized around estimands rather than around source systems.

Total deposit demand. Estimate D_{it}^* from internal transaction flows, firmographics, seasonal patterns, local industry conditions, rate regimes, credit conditions, and external business-density benchmarks. Federal Reserve rate and credit data enter as M_t and stress-state variables; Census and BLS data enter as industry-geography controls; transaction flows identify firm-specific operating liquidity.

Competitor share and local substitution. Estimate s_{ifpt} using focal-bank balances, branch footprint, competitor footprint, advertised rates, service attributes, branch-level market share, and bank-level balance-sheet controls. FDIC Summary of Deposits is useful here as a noisy local-market share anchor, while Call Reports and H.8/H.15 help measure funding and rate environments. These data cannot by themselves separate SME from consumer or wholesale share; the model must treat them as aggregate measurement equations.

Product incidence and size. Estimate product incidence and conditional dollar size from internal products, credit bureau and public-record features, SBA or public lending records where available, firmographics, and transaction proxies. For loans and credit lines, liens, UCC filings, property records, and credit bureau data are more direct than Fed aggregate data. For cards and merchant products, merchant settlement and payment-flow benchmarks are more relevant.

Source	Useful variables	Role in the model	Main limitation
Federal Reserve data portal	H.8, H.15, SLOOS, payments data, business-finance releases	Macro state, rate regime, credit conditions, payments context, stress scenario variables	Mostly aggregate, not SME wallet labels
FDIC Summary of Deposits	Branch deposits, institution and geography footprint	Local deposit market share and branch competition anchor	Mixes consumer, SME, commercial, and wholesale deposits
FDIC/FFIEC Call Reports	Bank-level deposits, loans, funding mix, uninsured deposit measures where reported	Competitor funding pressure, safety, pricing, and stress controls	Bank-level, not customer-level
Census CBP/BDS/BFS and BLS QCEW	Establishments, firm births, employment, wages, industry geography	Local SME universe, market size, growth, seasonality and industry stress	Establishment-level aggregates, not bank relationships
CRA, HMDA, and Section 1071 data	Small-business and mortgage lending geography, where available	Credit demand and market access context; future validation of small-business credit coverage [18, 19, 30]	Product scope and reporting definitions differ from deposit wallet
Commercial/vendor data	Firmographics, credit bureau, card/merchant/payroll, web and business registry data	Product incidence, size priors, external-wallet signals, prospect universe	Cost, coverage bias, consent, privacy, and model risk review
Consented or conversion data	External account balances, statement imports, ACH migration, payoff and transfer records	Direct validation of competitor wallet, move-over, cannibalization, and runoff	Sparse and selected sample

Table 1: Data sources for SME wallet estimation and their identification roles.

Move-over, cannibalization, and runoff. Estimate the movable fraction using data that reveal origin and displacement: account aggregation, conversion files, balance-transfer and payoff records, ACH/direct-deposit migration, customer onboarding statements, relationship- manager disposition codes, and randomized consolidation offers. This is the hardest part of the data problem. Public data can benchmark market size, but they cannot identify whether a given balance came from a competitor, from new liquidity demand, or from cannibalization of existing focal-bank balances.

Pricing and stress. Estimate pricing response from quoted rates, booked rates, fee waivers, rate-sheet changes, market rates, competitor advertised rates, and randomized or quasi-random offer variation. H.15, H.8, SLOOS, Call Reports, and FDIC market data provide the market and stress states. Internal balance decay and post-stress runoff are still required to value the deposits.

4.4 Practical Data Priorities

The most valuable next data investments are:

1. Build an account-level deposit and product panel with decision-time snapshots, so the model never uses post-offer information in scoring.
2. Create a transaction-feature mart for cash-flow volatility, payroll intensity, merchant settlement, ACH/wire counterparties, tax timing, and inferred external-bank activity.
3. Geocode customers, branches, and competitors; join FDIC Summary of Deposits, Census/BLS local business data, and competitor footprints.
4. Maintain a quote and offer ledger with offered rate, booked rate, rejected terms, fee waivers, exceptions, relationship-manager action, and eligibility rules.
5. Create validation samples through consented account aggregation, onboarding statement capture, conversion events, and relationship-manager migration codes.
6. Run small randomized or quasi-random pricing and consolidation tests to identify elasticity, move-over, cannibalization, and runoff.

Data governance is a first-order issue. A US bank must ensure that features, target definitions, offer decisions, and model monitoring comply with fair lending, privacy, consumer/commercial data-use restrictions, model-risk management, and customer consent requirements. The academic literature reviewed here does not by itself settle those compliance requirements.

For deployment, the modeling program should be governed as a bank model-risk and customer-decisioning system, not only as a marketing model. SR 11-7 requires conceptual soundness, ongoing monitoring, and independent validation for models used in bank decisioning [10]. Third-party data and fintech components should be reviewed under third-party risk-management expectations [11]. If the model is used in connection with small-business credit decisions or covered data collection, implementation should also be reviewed against the CFPB small-business lending rulemaking under Section 1071 [18].

5 Recommended Approach

The recommended approach is dynamic demand plus discrete choice. It begins by abandoning the idea that each SME has a fixed deposit wallet size. The bank should instead model two linked

objects:

Observed focal-bank deposits $_{ift}$ = Total SME liquidity demand $_{it}$ $\times$ Focal-bank deposit share $_{ift}$.

The first object is a dynamic balance-sheet and cash-flow problem. The second object is a bank-choice and deposit-share problem. A third object is the deposit amount that can move from competitors, which is generally smaller than the competitor-held amount. This decomposition is the central modeling recommendation of this survey.

5.1 Layer 1: Dynamic SME Liquidity Demand

The first layer estimates the SME's desired liquidity or total deposit demand:

$$D_{it}^* = g(X_{it}, M_t, R_{it}, C_{it}, B_{it}, \eta_i) + \varepsilon_{it}.$$

Here D_{it}^* is the firm's desired liquid balance, X_{it} includes firm-level cash-flow and transaction variables, M_t includes macro and market conditions, R_{it} includes relevant rates and fees, C_{it} includes financing conditions, B_{it} includes relationship variables, and η_i captures persistent firm heterogeneity. This layer should be estimated as a panel time-series, state-space, or hierarchical dynamic model. It should include seasonality, industry-by-time effects, geography-by-time effects, local market conditions, and relationship tenure. Pricing and stress extensions can be added after the baseline model is stable.

This layer answers questions such as: how much liquidity does the SME need now, how does that need vary with business scale and seasonality, and how much of a balance increase reflects normal cash-flow timing rather than wallet movement from a competitor?

5.2 Layer 2: Discrete Choice for Bank and Deposit Share

The second layer estimates how the SME allocates that liquidity and product demand across banks and outside options. This is where discrete choice demand estimation is useful. The alternative set must be declared before estimation. In the deposit use case, let $\mathcal{J}_{imt}^B$ denote modeled bank-relationship alternatives, let $\mathcal{J}_{imt}^{BP}$ denote bank-product alternatives when products are modeled separately, let $j = \text{res}$ denote residual unmodeled-bank balances, and let $j = 0$ denote true nonbank or outside liquidity such as money-market funds, cash-management vehicles, or no active movement. These categories should not be pooled unless the reporting question only requires total non-focal balances.

A simplified utility specification for SME i choosing alternative j in market m and time t is

$$U_{ijmt} = \beta_i' A_{jmt} - \alpha_i P_{ijmt} + \xi_{jmt} + \epsilon_{ijmt},$$

where A_{jmt} are observed bank/product attributes, P_{ijmt} is the SME-specific price or offer term, ξ_{jmt} is an unobserved product-market demand shock, and ϵ_{ijmt} is idiosyncratic taste. Alternatives may be defined as competing operating-account relationships or bank-product bundles, but the same model should not switch among these definitions without changing the share normalization and diversion interpretation.

Ma [46] is useful here because financial products often involve individual-specific or relationship-specific prices. In SME banking, effective prices may include negotiated deposit rates, treasury fees, minimum-balance requirements, fee waivers, and relationship pricing. If these prices vary systematically with firm size, risk, income, relationship tenure, or financing conditions, a standard BLP model that treats all customers in a market as facing the same product price can misestimate

price sensitivity. Ma’s two-step framework—first estimating individual-specific prices, then using a modified BLP-style demand model with selectivity correction—is therefore relevant for estimating SME choice among financial product alternatives.

Because Ma [46] is an unpublished working paper, it should be treated as a useful template for handling individual-specific financial-product prices rather than as settled banking practice. A bank should validate the two-step price and demand estimates against its own quote, booking, and campaign data before using them for pricing or targeting decisions.

For plain retail deposit products where all customers in a market truly face the same posted rate, a traditional BLP-style model may be sufficient. For SME relationship banking, however, prices and service terms are often negotiated or relationship-specific, so the individual-specific-price issue is likely material.

5.3 Layer 3: Product Size, Addressable Wallet, and Move-Over Amount

The third layer estimates dollars, balances, spend, commitments, or assets conditional on product need. For every product p , the model should separate incidence from severity:

$$\begin{aligned} r_{ipt}^* &= 1\{W_{ipt}^* > 0\}, \\ \Pr(r_{ipt}^* = 1 \mid h_{it}, X_{it}, M_t) &= \pi_p(h_{it}, X_{it}, M_t), \\ \log W_{ipt}^* \mid r_{ipt}^* = 1 &\sim F_p(\mu_p(h_{it}, X_{it}, M_t), \sigma_p^2). \end{aligned}$$

This two-part structure follows the limited-dependent-variable logic of Cragg [20]. It prevents a common mistake in wallet sizing: treating a product-propensity score as if it were a dollar balance. For products with highly skewed positive amounts, the model should output means, quantiles, and interval forecasts. If the severity model is fit in logs, dollar forecasts should use validated retransformation, such as smearing corrections, rather than naive exponentiation [25].

Given a draw of total product wallet and focal-bank share, the model produces the competitor-held amount

$$A_{ipt}^{-f(s)} = \max\{W_{ipt}^{*(s)} - a_{ifpt}^{\text{obs}}, 0\}.$$

If the bank needs bank-specific competitor estimates, this amount can be allocated across competitors using the discrete-choice shares:

$$a_{ibpt}^{(s)} = s_{ibpt}^{(s)} A_{ipt}^{-f(s)} / \sum_{c \neq f} s_{icpt}^{(s)}, \quad b \neq f.$$

This allocation is only over modeled banks. If important banks are not modeled, include a residual unmodeled-bank category in the denominator and report it separately. A true nonbank outside option should be allocated dollars only when the wallet definition includes nonbank liquidity; otherwise it belongs in the share model but not in bank-specific competitor balances.

The move-over amount is a separate estimand. For offer o and horizon H , write

$$\Delta a_{iop}^{\text{move}}(H) = M_{iop}(H) A_{ipt}^{-f}, \quad 0 \leq M_{iop}(H) \leq 1.$$

The fractional response $M_{iop}(H)$ can be modeled with a fractional-logit, beta-regression, hierarchical Bayesian, or causal-forest layer using offer terms, switching costs, relationship depth, competitor attributes, price gaps, and product frictions. It must be estimated from data that observe or identify movement: conversion events, balance-transfer or payoff records, account aggregation before and after onboarding, ACH or wire migration patterns, relationship migrations, merger/acquisition integrations, or randomized offer cells.

Observed focal-bank balance growth is not itself the move-over amount. The accounting identity should be

$$\Delta a_{ifp,t:t+H}^{\text{obs}}(o) = \Delta a_{iop}^{\text{move}}(H) + \Delta a_{iop}^{\text{expand}}(H) - \Delta a_{iop}^{\text{cann}}(H) + u_{iopH},$$

where Δa^{move} is migrated competitor wallet, Δa^{expand} is expansion of the SME's total demand for product p , and Δa^{cann} is cannibalization of existing focal-bank balances or organic growth. Thus a deposit initiative can create large observed growth because the firm expanded, because it moved competitor balances, or because it shifted value from existing focal-bank balances. These components have different profitability, funding, liquidity, and credit implications.

5.4 Layer 4: Counterfactual Offer and Wallet-Share Simulation

Once the demand, choice, and amount layers are estimated, the bank can run counterfactuals. For a candidate deposit action o ,

$$\widehat{D}_{ifo,t+1} = \widehat{D}_{i,t+1}^* \times \widehat{s}_{ifo,t+1},$$

where $\widehat{D}_{i,t+1}^*$ comes from the dynamic liquidity-demand layer and $\widehat{s}_{ifo,t+1}$ comes from the discrete-choice share layer. This lets the bank distinguish three cases that look similar in a static wallet score:

1. The SME has high total liquidity demand, but the focal bank has low share because the bank's price, service, or relationship position is uncompetitive.
2. The SME has low current liquidity demand, but high future demand under seasonal growth.
3. The SME has high observed balances at the focal bank, but low incremental response to marketing because it would have stayed anyway.

The discrete-choice layer can also quantify willingness to pay for attributes: branch access, relationship-manager quality, digital onboarding, faster credit decisions, integrated treasury tools, deposit-rate premium, fee waivers, or onboarding support. These willingness-to-pay estimates are useful for evaluating deposit capture actions rather than only ranking prospects.

5.5 Layer 5: Causal Uplift and Risk-Adjusted Decisioning

The dynamic demand and discrete-choice layers estimate demand and substitution. They do not by themselves prove that a marketing action causes incremental balances. The final layer should use randomized holdouts or credible quasi-experiments to estimate incremental response:

$$\tau_{io}^Y = \mathbb{E}[Y_i(o) - Y_i(0) \mid \mathcal{I}_{it}].$$

The superscript matters: τ_{io}^D is a deposit-balance contrast, τ_{io}^{move} is migrated competitor wallet, and τ_{io}^V is risk-adjusted value. They need not have the same sign or the same identification support. The recommended targeting criterion is risk-adjusted incremental value:

$$\text{NPV}_{io} = m_D \widehat{\Delta D}_{io} + \sum_p m_p \widehat{\Delta a}_{iop} - c_o - \lambda \widehat{\text{Risk}}_{io},$$

where $\widehat{\Delta D}_{io}$ is incremental deposit balance, m_D is the deposit margin or funding value, m_p are product margins, c_o is offer cost, and the risk term captures credit, liquidity, conduct, and compliance constraints.

Thus the recommended system is:

Dynamic liquidity demand + discrete choice deposit share + product amount and move-over + causal uplift + risk-adjusted optimization.

This framework keeps the core economic distinction clear: deposit balances are dynamic demand, while wallet capture is a competitive choice and response problem.

5.6 Summary of the Recommended Model

The recommended model combines the surveyed architectures into one generative-and-decision system. This subsection gives the compact version; the next two subsections state the estimands, estimator interfaces, and validation gates. The key notation is:

Symbol	Meaning
W_{ipt}^*	Latent total wallet for product p
A_{ipt}^{-f}	Competitor-bank-held amount, excluding true nonbank outside liquidity unless stated
D_{it}^*	Latent total SME deposit demand
$s_{ifpt}(o)$	Focal-bank share of product p under offer o
$M_{iop}(H)$	Structural fraction of competitor-held product p that moves under offer o
$\mu_{iop}(H)$	Expected moved dollars
τ_{io}^Y	Causal contrast for explicitly named outcome Y
$J_{io}(H)$	Risk-adjusted incremental relationship value

Let h_{it} denote a latent SME state containing firm scale, liquidity need, credit demand, relationship preference, price sensitivity, and switching cost. The state evolves over time:

$$h_{it} = F(h_{i,t-1}, X_{it}, M_t, L_{it}, \omega_{it}),$$

where X_{it} are firm signals, M_t are market conditions, L_{it} captures financing conditions and liquidity constraints, and ω_{it} is an innovation. Deposit need and total product wallet are generated from this state:

$$W_{ipt}^* \sim p(W_{ipt}^* | h_{it}, X_{it}, M_t), \quad y_{ipt}^* \sim p(y_{ipt}^* | h_{it}, W_{ipt}^*).$$

For deposits, the product-wallet equation should be specialized as dynamic liquidity demand:

$$D_{it}^*(M_t) = g(h_{it}, \text{cash-flow risk}_{it}, M_t, \text{relationship}_{it}).$$

The allocation of that demand across the focal bank, competitors, and outside options is then governed by a discrete-choice share model. Let $\mathcal{B}_{mt}^{\text{mod}}$ be the modeled bank set, let $j = \text{res}$ denote residual unmodeled-bank balances, and let $j = 0$ denote true nonbank outside liquidity. Then

$$s_{ibpt}(o) = q_{bp}(A_{bpt}, P_{ibpt}(o, M_t), B_{ibt}, h_{it}, M_t, \xi_{bpt}), \quad a_{ibpt}(o, M_t) = s_{ibpt}(o, M_t) W_{ipt}^*(M_t).$$

Thus deposit pricing and stress enter through both total demand $D_{it}^*(M_t)$ and captured share $s_{ifpt}(o, M_t)$. The competitor-held and addressable components are

$$A_{ipt}^{-f}(0) = \sum_{b \in \mathcal{B}_{mt}^{\text{mod}}, b \neq f} a_{ibpt}(0) + a_{i,\text{res},pt}(0), \quad A_{ipo}^{\text{addr}}(H) = M_{ipo}(H) A_{ipt}^{-f}(0).$$

The nonbank outside option $j = 0$ is included in A_{ipt}^{-f} only if the business definition of wallet includes nonbank liquidity. Otherwise it affects choice normalization but is reported separately from competitor-bank wallet. The causal move-over amount satisfies the accounting decomposition

$$\Delta a_{ifp,t:t+H}(o) = \Delta a_{iop}^{\text{move}}(H) + \Delta a_{iop}^{\text{expand}}(H) - \Delta a_{iop}^{\text{cann}}(H),$$

where only the first term is migrated competitor wallet. The second term is growth in total product demand, and the third term is cannibalization of existing focal-bank balances. Finally, campaign experiments or quasi-experiments estimate the incremental effect of taking action o on explicitly named outcomes. For risk-adjusted relationship value,

$$\tau_{io}^V = \mathbb{E}\{V_i(o) - V_i(0) \mid h_{it}, X_{it}\},$$

where $V_i(o)$ is risk-adjusted relationship value. This is distinct from τ_{io}^D for deposit balances and τ_{io}^{move} for migrated competitor dollars. The decision rule is to choose offers that maximize expected incremental value subject to eligibility, fairness, liquidity, risk, capital, and relationship-manager capacity constraints.

This combined model is more complete than any single surveyed architecture. Latent wallet reconstruction fills the missing competitor-balance problem. Dynamic liquidity demand prevents the deposit target from being treated as a fixed stock. Two-part amount modeling estimates the size of each product conditional on ownership. Discrete choice supplies the substitution and pricing mechanism needed for counterfactual deposit actions. Section 6 extends this same demand-and-share decomposition to stress-condition pricing, rate sensitivity, and runoff. Move-over modeling distinguishes competitor balances that exist from balances that can be captured. Causal uplift prevents the bank from confusing high propensity with incremental response. Relationship-lending research supplies the reason to optimize the deposit relationship rather than the deposit margin in isolation.

5.7 Why This Approach Is Preferable: Estimands and Losses

The practical decision is not to estimate a scalar “wallet size.” It is to choose an offer o from a feasible set $\mathcal{O}_{it}$ using information $\mathcal{I}_{it}$ available at decision time. A rigorous target is the conditional incremental relationship value over horizon H :

$$J_{io}(H) = \mathbb{E}[V_{i,t+H}(o) - V_{i,t+H}(0) \mid \mathcal{I}_{it}, o \in \mathcal{O}_{it}],$$

where V includes deposit funding value, product margin, servicing cost, credit loss, liquidity stability, incentive cost, and compliance constraints. The policy target is

$$\pi^*(\mathcal{I}_{it}) = \arg \max_{o \in \mathcal{O}_{it}} J_{io}(H) \quad \text{subject to risk, fairness, capacity, and eligibility constraints.}$$

A score is decision-relevant only to the extent that it is monotone in $J_{io}(H)$ for feasible offers. A fixed wallet-size score generally is not, because it omits dynamic liquidity demand, competitive substitution, treatment response, product-specific amount distributions, movable competitor wallet, and risk-adjusted value.

For deposits, the decomposition makes the failure precise. Let $D_{ift}(o) = s_{ift}(o)D_{it}^*(o)$ denote the focal-bank deposit balance under offer o , where D_{it}^* is total SME liquidity demand and s_{ift} is the focal bank’s share. Relative to no action,

$$\begin{aligned} \Delta D_{if,t+H}(o) &= D_{i,t+H}^*(0)\Delta s_{if,t+H}(o) + s_{if,t+H}(0)\Delta D_{i,t+H}^*(o) \\ &\quad + \Delta s_{if,t+H}(o)\Delta D_{i,t+H}^*(o), \end{aligned}$$

where $\Delta s_{if,t+H}(o) = s_{if,t+H}(o) - s_{if,t+H}(0)$ and $\Delta D_{i,t+H}^*(o) = D_{i,t+H}^*(o) - D_{i,t+H}^*(0)$. A static wallet model may forecast net focal-bank balance changes, but it does not separately identify these channels. In particular, it cannot tell whether the observed change came from share capture,

changes in the firm’s total liquidity demand, the interaction of the two, or causal incrementality of the offer without additional structure or experimental variation.

For non-deposit products the same issue appears as an incidence-size-movement decomposition. Let $r_{ipt}^* = 1\{W_{ipt}^* > 0\}$ and let $M_{iop}(H)$ denote the movable fraction of competitor-held product p . The gross move-over target is

$$G_{iop}(H) = \mathbb{E}\left[1\{r_{ipt}^* = 1\}A_{ipt}^{-f}M_{iop}(H) \mid \mathcal{I}_{it}, o\right].$$

This formula shows why three models are required: product incidence, conditional amount, and offer-specific movable fraction. Estimating only the first factor answers “who may need the product,” not “how many dollars are at competitors” or “how many dollars are likely to move.” A factorized approximation can be used for transparency, but it should be validated because competitor-held amount and movable fraction are likely correlated through firm size, switching costs, relationship depth, and competitor lock-in.

The fixed-wallet approach is valid only under strong assumptions:

1. $D_{i,t+H}^*(o) = D_{it}^*$, so the firm’s liquidity demand is fixed over the horizon and unaffected by market conditions or the offer.
2. $s_{if,t+H}(o)$ depends only on an observed propensity score, not on unobserved competitor attributes, negotiated prices, switching frictions, or relationship-manager actions.
3. Product incidence is enough to determine dollar size, so conditional amount distributions and retransformation uncertainty can be ignored.
4. The movable fraction $M_{ipo}(H)$ is either one or a known constant, so all competitor-held balances are treated as equally addressable.
5. The offer assignment mechanism is ignorable: $Y_i(o) \perp O_i \mid \mathcal{I}_{it}$, or there is randomized or quasi-random variation that identifies the counterfactual.
6. Product margins, balance stability, credit risk, and servicing cost are either constant across SMEs or irrelevant to the decision.
7. Moving deposits does not substitute away from other profitable products or cannibalize existing focal-bank balances.

The literatures surveyed above weaken these assumptions: corporate liquidity research rejects time-invariant cash demand [2, 4, 7, 53]; structural demand models treat price and product attributes as drivers of share [9, 23]; causal-response methods separate propensity from treatment effect [6, 45, 66]; and relationship-lending research implies that deposits, monitoring, and credit products jointly affect value [8, 48, 52, 54].

5.8 How the Literature Components Fit Together: Estimation and Interfaces

The architecture should be written as a set of estimators with explicit interfaces. Let

$$\mathcal{I}_{it} = \sigma\{X_{i,1:t}, M_{1:t}, A_{1:t}, P_{i,1:t}^{\text{obs}}, a_{if,1:t}^{\text{obs}}, Y_{i,1:t}^{\text{obs}}, O_{i,1:t-1}\}$$

denote the information available before the period- t decision. The system must not use information realized after the offer when estimating a decision score. Let $\mathcal{A}_{it}$ denote anchoring data, such as account-aggregation records, post-conversion balances, customer self-reports, bureau or public

Model class	Estimand it targets	Failure mode for SME deposits	Role in recommended system
Fixed wallet score	Rank or point estimate of total balances	Assumes liquidity demand and wallet are stable; confounds size and share	Baseline comparator only
Cross-sell/NPTB	Product incidence or next-product probability	Predicts need, not balance size or causal response	Product graph and incidence prior
Two-part amount model	W_{ipt}^* , A_{ipt}^{-f} , and conditional product size	Needs anchors and careful dollar retransformation	Product-size and competitor-wallet layer
Dynamic deposit model	D_{it}^* or D_{ift} over time	Forecasts balances but not competitive substitution by itself	Liquidity-demand layer
Discrete choice	$s_{ibpt}(o)$ and substitution elasticities	Needs prices, attributes, and identification of price effects	Deposit-share and offer-simulation layer
Move-over/uplift	$M_{ipo}(H)$, $\mu_{iop}(H)$, $J_{io}(H)$, or τ_{io}^Y under an action	Needs migration evidence and randomized or credible quasi-random variation	Campaign-response and policy layer

records, branch-level market shares, and market-cell totals. These anchors are essential because internal balances alone do not identify competitor balances. Branch-level market share is informative but mixed: it usually combines consumer, SME, commercial, and wholesale deposits. It should therefore enter as an aggregate constraint or noisy measurement, not as a direct SME label.

Module A: latent wallet and state filtering. The target is the filtering distribution

$$F_{it}^{W,h} = p(h_{it}, \{W_{ipt}^*, A_{ipt}^{-f}\}_p, D_{it}^* \mid \mathcal{I}_{it}, \mathcal{A}_{it}),$$

not a point estimate. A useful measurement system is

$$\begin{aligned}
r_{ipt}^* &= 1\{W_{ipt}^* > 0\}, \\
\log W_{ipt}^* \mid r_{ipt}^* = 1 &= \mu_p(h_{it}, X_{it}, M_t) + u_{ipt}, \\
a_{ifpt}^{\text{obs}} &= s_{ifpt} W_{ipt}^* + \varepsilon_{ifpt}^a, \\
A_{ipt}^{\text{anchor}} &= W_{ipt}^* + \varepsilon_{ipt}^A \quad \text{when an anchor is observed,} \\
S_{fmt}^{\text{branch}} &= \sum_g \omega_{gmt} S_{fgmt} + \varepsilon_{fmt}^S.
\end{aligned}$$

The anchor equations are what provide level information for the latent wallet. Without individual anchors, branch-market aggregates, or credible market-cell restrictions, the model may rank SMEs but cannot claim point identification of total competitor wallet. Mixed branch aggregates should be treated as noisy constraints, not as direct SME wallet observations. The estimator can be maximum likelihood with EM, Bayesian data augmentation, or a mixed-data factor model [21, 37, 38]. For product amounts, the incidence and positive-size equations should be estimated and validated separately, as in two-part limited dependent variable models [20]. Its required output is a set of draws

$$\{h_{it}^{(s)}, W_{ipt}^{*(s)}, A_{ipt}^{-f(s)}, D_{it}^{*(s)}\}_{s=1}^S,$$

which preserves wallet uncertainty for downstream choice and policy simulation.

Module B: dynamic deposit demand. For deposits, write $d_{it}^* = \log D_{it}^*$. A minimal panel state equation is

$$d_{it}^* = \rho d_{i,t-1}^* + \beta' X_{it} + \gamma' M_t + \lambda_{g(i),t} + \alpha_i + \eta_{it},$$

where $\lambda_{g(i),t}$ is an industry- or geography-by-time effect and α_i is persistent firm heterogeneity. The focal-bank observation is a measurement equation:

$$\log a_{if,\text{dep},t}^{\text{obs}} = d_{it}^* + \log s_{if,\text{dep},t} + \varepsilon_{it}^D.$$

This equation shows the identification problem directly: observed focal-bank deposits identify $d_{it}^* + \log s_{if,t}$, not its two components, unless the model has share anchors, conversion observations, account-aggregation data, or credible restrictions on $s_{if,t}$. Dynamic-panel estimators, hierarchical state-space models, and stress-regime interactions are then estimation choices; the statistical interface is the predictive distribution

$$p(D_{i,t+h}^* \mid \mathcal{I}_{it}, \mathcal{A}_{it}), \quad h = 1, \dots, H.$$

Arellano-Bond-style dynamic panels are useful when many SMEs are observed for a moderate number of periods and lagged balances interact with firm effects [5].

Module C: discrete choice and deposit share. Given draws of h_{it} and D_{it}^* , the choice module estimates how wallet is allocated across the focal bank, competitors, residual unmodeled banks, and true outside options. In a bank-level deposit model, j indexes bank relationships plus residual and outside alternatives. In a bank-product model, j indexes bank-product bundles. The interpretation of shares and diversion must follow that choice-set definition. For alternative j in market m ,

$$U_{ijmt}(o) = \delta_{jmt} + \beta_i' A_{ijmt} - \alpha_i P_{ijmt}(o) + \kappa_i' B_{ijmt} + \xi_{jmt} + \epsilon_{ijmt},$$

so that

$$\Pr(c_{it} = j \mid o, \mathcal{I}_{it}) = \int \frac{\exp(V_{ijmt}(o; \nu))}{\sum_{\ell \in \mathcal{J}_{imt}} \exp(V_{i\ell mt}(o; \nu))} dF_\nu(\nu).$$

The focal-bank product share is

$$s_{ifpt}(o) = \sum_{j \in \mathcal{J}_{fpt}} \Pr(c_{it} = j \mid o, \mathcal{I}_{it}).$$

BLP-style models supply the counterfactual substitution structure [9]; simulation methods are standard for heterogeneous choice models [60]. Ma's framework is relevant when P_{ijmt} is individual-specific and unchosen prices must be inferred or selection-corrected [46]. The module must output not only $s_{ifpt}(o)$ but also the diversion matrix

$$\mathcal{D}_{jk}(x) = -\frac{\partial s_k(x)/\partial P_j}{\partial s_j(x)/\partial P_j},$$

where x denotes the segment, market, or individual state at which diversion is evaluated. Because deposit capture depends on where balances are diverted from, the diversion matrix should add up over the same alternative set used in the choice model. In the main deposit use case, it is used to distinguish balances captured from modeled competitors, residual unmodeled banks, true outside options, and balances shifted within the focal bank.

Module D: move-over amount and causal response. The choice model predicts substitution under specified prices and attributes; it does not by itself identify the effect of marketing contact or relationship-manager action. For an offer o , define

$$e_o(x) = \Pr(O_i = o \mid \mathcal{I}_{it} = x), \quad m_o(x) = \mathbb{E}[V_i \mid O_i = o, \mathcal{I}_{it} = x].$$

Under consistency, overlap, and randomized or conditionally ignorable assignment given $\mathcal{I}_{it}$, an orthogonal score for the value difference between offer o and no offer is

$$\begin{aligned}\psi_i(o) &= m_o(X_i) - m_0(X_i) + \frac{1\{O_i = o\}}{e_o(X_i)} \{V_i - m_o(X_i)\} \\ &\quad - \frac{1\{O_i = 0\}}{e_0(X_i)} \{V_i - m_0(X_i)\}.\end{aligned}$$

This score is a debiased estimating equation for an average value contrast, or for a segment contrast after pre-specified conditioning. Turning it into individualized uplift requires a separate heterogeneous-effect step, such as a causal forest or an orthogonalized regression of the score on the decision-time state, and that step should be validated as a policy rule rather than treated as automatic identification of customer-level treatment effects. This connects classical propensity-score logic [58], uplift modeling [45], causal forests [66], and orthogonal/double machine learning [17]. The module outputs a stated estimand—average lift for a segment, conditional lift $\hat{\tau}_o^Y(x)$, or an individualized policy score $\hat{J}_{io}$ —with uncertainty, overlap diagnostics, and a flag indicating whether the estimate is experimental, quasi-experimental, or only associational.

For product-size decisions, the outcome vector should decompose the observed change:

$$Y_{iopH} = \left(\Delta a_{iop}^{\text{move}}, \Delta a_{iop}^{\text{expand}}, \Delta a_{iop}^{\text{cann}}, \Delta \text{NPV}_{iop} \right).$$

The move-over estimand is

$$\mu_{iop}(H) = \mathbb{E}[\Delta a_{iop}^{\text{move}}(H) \mid \mathcal{I}_{it}, o], \quad \tilde{m}_{iop}(H) = \frac{\mu_{iop}(H)}{\mathbb{E}[A_{ipt}^{-f} \mid \mathcal{I}_{it}]},$$

where $\tilde{m}_{iop}(H)$ is an operational reporting ratio, not generally the same object as the structural fraction $M_{iop}(H)$. A ratio of posterior means need not equal the posterior mean of a latent fraction under heterogeneity. The ratio should be truncated to $[0, 1]$ only as a reporting convention and only when the denominator is identified. When data do not separately observe migrated competitor balances, the model should report net incremental balance, not label it move-over. The required training data are events that identify origin or displacement: account-transfer records, payoff of competitor debt, balance-transfer activity, pre/post account aggregation, direct-deposit or ACH migration, or randomized offers with follow-up account consolidation evidence.

Module E: policy optimization and uncertainty propagation. The policy layer consumes simulation draws, not point predictions:

$$\left\{ D_{i,t+h}^{*(s)}, W_{ip,t+h}^{*(s)}, A_{ip,t+h}^{-f(s)}, s_{ifp,t+h}^{(s)}(o), M_{iop}^{(s)}(H), \Delta a_{iop}^{\text{move}(s)}, \tau_{io}^{Y,(s)}, R_{io}^{(s)} \right\}_{s=1}^S.$$

For each feasible offer,

$$\hat{J}_{io}(H) = \frac{1}{S} \sum_{s=1}^S \left[V_{i,t+H}^{(s)}(o) - V_{i,t+H}^{(s)}(0) \right].$$

The recommended action is

$$\hat{\pi}(\mathcal{I}_{it}) = \arg \max_{o \in \mathcal{O}_{it}} \hat{J}_{io}(H)$$

subject to chance and resource constraints such as

$$\Pr(R_{io} > r_{\max} \mid \mathcal{I}_{it}) \leq \alpha_R, \quad \sum_i 1\{\hat{\pi}(\mathcal{I}_{it}) = o\} \leq C_o.$$

The required policy output is the chosen offer, expected incremental value, credible interval, second-best action, margin of victory, and binding constraints. If the credible interval for the margin of victory includes zero, the system should recommend experimentation or relationship-manager review rather than automatic targeting.

5.9 Potential Issues and Remedies: Identification Gates

The model should pass explicit gates before it is used for material pricing or targeting. Each gate has a null failure mode, a diagnostic statistic, and a permitted action if the gate fails.

Gate 1: latent wallet scale. Null failure mode: the model has learned relative customer size but not the level of competitor wallet. Let $\mathcal{V}_{gp}$ be an external validation set for product p and segment g . Define relative bias and interval coverage:

$$B_{gp} = \frac{1}{|\mathcal{V}_{gp}|} \sum_{i \in \mathcal{V}_{gp}} \frac{\widehat{W}_{ipt}^* - W_{ipt}^A}{W_{ipt}^A}, \quad C_{gp}(\alpha) = \frac{1}{|\mathcal{V}_{gp}|} \sum_{i \in \mathcal{V}_{gp}} 1\{W_{ipt}^A \in I_{ipt}^{1-\alpha}\}.$$

The gate passes only if $|B_{gp}| \leq b_{\max}$ and $|C_{gp}(\alpha) - (1 - \alpha)| \leq c_{\max}$ for material segments. If it fails, the model may be used for ranking or exploratory relationship planning, but not for dollar wallet claims. The remedy is more anchor data, a simpler segment model, or a sensitivity interval for W_{ipt}^* [21, 37].

Gate 2: missing-not-at-random competitor wallet. Null failure mode: missing competitor information is systematically related to the unobserved wallet. One sensitivity model is

$$\text{logit Pr}(R_{ipt}^A = 1 \mid X_{it}, W_{ipt}^*) = \zeta' X_{it} + \lambda \log W_{ipt}^*,$$

where R_{ipt}^A indicates anchor observation and λ controls selection on the latent wallet. The diagnostic is the policy range

$$\Delta_\Lambda = \sup_{\lambda \in \Lambda} \widehat{J}_{io}^{(\lambda)} - \inf_{\lambda \in \Lambda} \widehat{J}_{io}^{(\lambda)}.$$

If the recommended action changes for plausible λ , the action is not identified from the available data. The remedy is to collect validation anchors or restrict the deployment to segments where the policy is stable over Λ .

Gate 3: price selection. Null failure mode: unchosen prices are missing selectively. Let R_{ijmt}^P indicate that the effective price for alternative j is observed. A selection correction requires variables Z_{ijmt}^P that affect price observation or quote generation but do not directly enter demand after conditioning on $\mathcal{I}_{it}$:

$$R_{ijmt}^P = 1\{\pi' Z_{ijmt}^P + v_{ijmt} > 0\}.$$

The diagnostic is whether R_{ijmt}^P remains predictable from risk, size, industry, or relationship variables after the proposed correction. If no credible exclusion or randomized quote variation exists, price elasticities should be labeled associational. Ma's selectivity correction and Heckman's sample-selection framework give the relevant templates [33, 46].

Gate 4: endogenous prices and offers. Null failure mode: $E[C_{ijmt}\xi_{jmt}] \neq 0$, so estimated price sensitivity loads on unobserved demand or unobserved quality. With instruments Z_{jmt} , the core moment is

$$\mathbb{E}[Z_{jmt}\xi_{jmt}(\theta)] = 0.$$

Diagnostics include first-stage strength, sign stability under alternative instrument sets, placebo effects before rate-sheet changes, and comparison with randomized offer cells. If the instrument set fails, the model may simulate attribute preference but should not be used for rate optimization. Remedies are randomized price tests, cost or rate-sheet instruments, policy-threshold variation, or a reduced-form campaign experiment [9, 23].

Gate 5: dynamic forecast validity. Null failure mode: the deposit model confuses persistence, seasonality, and wallet growth. Because D_{it}^* is latent for most SMEs, the gate must be defined on an observable implication unless an anchored validation sample is available. For horizon h , compare the model loss with a transparent baseline class $\mathcal{B}$:

$$R_h = \frac{\sum_{(i,t) \in \mathcal{T}_{\text{test}}} L(a_{if,\text{dep},t+h}^{\text{obs}}, \hat{D}_{i,t+h}^*, \hat{s}_{if,\text{dep},t+h})}{\min_{b \in \mathcal{B}} \sum_{(i,t) \in \mathcal{T}_{\text{test}}} L(a_{if,\text{dep},t+h}^{\text{obs}}, \hat{a}_{if,\text{dep},t+h}^b)}.$$

For anchored samples with observed or credible proxy total demand $D_{i,t+h}^A$, a separate latent-demand loss $L(D_{i,t+h}^A, \hat{D}_{i,t+h}^*)$ should be reported. The two losses answer different questions and should not be pooled. The gate requires $R_h < 1$ for the horizons used in decisioning, calibrated prediction intervals, no material residual autocorrelation, and separate stress-window performance. If it fails, use the dynamic model only as a feature generator and fall back to the best transparent baseline. Dynamic-panel tests and lagged dependent variable methods are relevant here [5].

Gate 6: substitution and cannibalization. Null failure mode: the choice model predicts adoption but not where balances come from. The diversion matrix

$$\mathcal{D}_{jk} = -\frac{\partial s_k / \partial P_j}{\partial s_j / \partial P_j}$$

should have economically plausible signs and satisfy adding-up over the inside and outside alternatives. The gate also requires post-offer accounting:

$$\Delta a_{ifp}^{\text{net}} = \Delta a_{ifp}^{\text{new}} - \Delta a_{ifp}^{\text{cannibalized}}.$$

If diversion or cannibalization cannot be measured, the model may rank deposit interest but should not forecast incremental captured balances. Remedies include nested choice sets, random coefficients, explicit outside options, and deposit offer experiments [9, 60].

Gate 7: move-over amount identification. Null failure mode: the model observes focal-bank growth after an offer and mislabels it as competitor wallet moved over. Let Z_{iopH}^{mig} denote migration-origin evidence, such as account transfer records, payoff of competitor debt, balance-transfer activity, account-aggregation before and after conversion, or direct-deposit and ACH rerouting. The measurement equation is

$$Z_{iopH}^{\text{mig}} = \Delta a_{iop}^{\text{move}}(H) + \epsilon_{iopH}^{\text{mig}},$$

when such evidence is observed. The gate requires the reconciled decomposition

$$\Delta a_{ifp,t:t+H}^{\text{obs}} = \hat{\Delta a}_{iop}^{\text{move}} + \hat{\Delta a}_{iop}^{\text{expand}} - \hat{\Delta a}_{iop}^{\text{cann}} + \hat{u}_{iopH}$$

to have small residuals in validation samples, economically plausible signs, and the cap

$$0 \leq \widehat{\Delta a_{iop}^{\text{move}}} \leq \widehat{A_{ipt}^{-f}}$$

except for measured new-product demand or measurement-error allowances. If this gate fails, the bank may report net incremental balances but should not report “dollars moved from competitors.” Remedies are explicit migration questions in onboarding, account-aggregation validation samples, balance-transfer and payoff tracking, relationship-manager disposition codes, and randomized consolidation offers.

Gate 8: causal incrementality. Null failure mode: high predicted propensity is mistaken for treatment effect. For each offer, require overlap

$$\epsilon < e_o(\mathcal{I}_{it}) < 1 - \epsilon$$

on the deployment population, standardized mean differences below a chosen threshold after weighting or matching, no placebo effect on pre-treatment outcomes, and decision-relevant lift estimates in randomized holdouts with calibrated uncertainty. The requirement is credible identification and calibration, not uniformly positive lift in every subgroup. If the gate fails, the output must be labeled propensity or association, not uplift. Randomized holdouts are the preferred remedy; when they are unavailable, use propensity score adjustment, causal forests, or orthogonal scores with explicit overlap and placebo diagnostics [17, 45, 58, 66].

Gate 9: risk-adjusted policy validity. Null failure mode: the model maximizes balance while violating profitability, liquidity, credit, conduct, or fairness constraints. The policy must satisfy constraints such as

$$\mathbb{E}[\text{NPV}_{io} \mid \mathcal{I}_{it}] > 0, \quad \Pr(\text{Runoff}_{io} > r_{\max} \mid \mathcal{I}_{it}) \leq \alpha_R, \quad |\text{Calib}_g - \text{Calib}_{g'}| \leq \delta$$

for monitored groups g, g' , where Calib_g is the chosen calibration error for group g on the governed target, such as predicted incremental value, approval probability, balance lift, or adverse outcome rate. Diagnostics include balance decay, runoff under stress, credit migration, complaint rates, subgroup calibration, and relationship-manager override patterns. If this gate fails, targeting should be limited to manual review or randomized learning cells until the binding risk is understood.

6 Deposit Pricing and Stress-Condition Demand Extension

The recommended model in Section 5 gives the baseline architecture:

$$D_{ift}(o, M_t) = D_{it}^*(M_t) s_{ift}(o, M_t).$$

This section extends that architecture for deposit pricing and stress conditions. The extension is not a separate business question. It answers the same wallet-capture question under a richer environment: given market condition M_t , stress state R_t , and deposit action o , how much total SME deposit demand exists, what share can the focal bank capture, what price is required, and how stable is the captured balance? The stress-pricing problem is therefore a joint problem of *quantity*, *share*, *price*, and *duration*. A high offered rate is attractive only if the resulting balances are incremental, sufficiently stable, and valuable after funding, liquidity, cannibalization, and relationship costs.

6.1 Pricing Objective and Decision Estimands

Let r_{io} denote the effective deposit rate, earnings credit, fee waiver, or relationship-pricing term embedded in action o . A pricing decision should maximize incremental relationship value, not gross balance:

$$J_{io}^{\text{price}}(H) = \mathbb{E} \left[\sum_{h=1}^H \delta^h \{ v_{t+h}^F D_{if,t+h}(o, R_{t+h}) - c_{io,t+h}^r D_{if,t+h}(o, R_{t+h}) \right. \\ \left. + \Pi_{i,t+h}^{\text{rel}}(o) - \text{EL}_{i,t+h}(o) - \text{LC}_{i,t+h}(o) \} - C_{io} \mid \mathcal{I}_{it} \right],$$

where v_{t+h}^F is the funding or liquidity value of deposits, c_{io}^r is the rate or pricing cost, Π^{rel} is non-deposit relationship profit, EL is expected credit loss, LC is liquidity or runoff cost, and C_{io} is campaign or onboarding cost. The decision rule is

$$o_i^* = \arg \max_{o \in \mathcal{O}_{it}} J_{io}^{\text{price}}(H)$$

subject to conduct, fair-lending, liquidity, concentration, relationship-manager capacity, and profitability constraints. This prevents the pricing model from recommending expensive balances that are transient, risky, or cannibalized from existing focal-bank deposits.

The target should be an *incremental* pricing estimand. Let

$$\Delta D_{ifo,t+h}^r = D_{if,t+h}^r(o) - D_{if,t+h}^r(0)$$

denote incremental focal-bank deposits in scenario r . The object to be priced is not this total alone, but the decomposition

$$\Delta D_{ifo,t+h}^r = \Delta D_{ifo,t+h}^{\text{move},r} + \Delta D_{ifo,t+h}^{\text{expand},r} - \Delta D_{ifo,t+h}^{\text{cann},r} + \epsilon_{io,t+h}^r.$$

The first term is competitor wallet moved to the focal bank, the second is growth in the SME's total liquidity demand, and the third is cannibalization of existing focal-bank balances or organic inflows. Stress pricing should report all three terms. Otherwise, the bank may pay a premium rate for balances it already would have retained or for temporary balances that leave as soon as market conditions normalize.

6.2 Economic Pricing Logic and Marginal Value

A useful way to discipline the pricing decision is to write the period contribution margin as

$$\pi_{io,t+h}^r = v_{t+h}^{F,r} D_{if,t+h}^r(o) - r_{io,t+h} D_{if,t+h}^r(o) - \Phi_{io,t+h}^r(D_{if,t+h}^r(o)) + \Pi_{io,t+h}^{\text{rel},r},$$

where Φ includes servicing cost, incentive cost, liquidity charge, operational cost, and runoff or concentration penalties. Holding the feasible offer class fixed, an interior rate choice must satisfy

$$0 = \mathbb{E} \left[\sum_{h=1}^H \delta^h \left\{ (v_{t+h}^{F,r} - r_{io,t+h}) \frac{\partial D_{if,t+h}^r}{\partial r_{io}} - D_{if,t+h}^r \right. \right. \\ \left. \left. - \frac{\partial \Phi_{io,t+h}^r}{\partial r_{io}} + \frac{\partial \Pi_{io,t+h}^{\text{rel},r}}{\partial r_{io}} \right\} \mid \mathcal{I}_{it} \right].$$

This condition is useful for management because it separates the *marginal balance effect* from the *infra-marginal cost*: raising the rate may attract or retain deposits, but the higher rate is usually paid on the full eligible balance, not only on the marginal dollars. In stress conditions this tradeoff becomes sharper because $v^{F,r}$, runoff cost, and competitor price responses can change at the same time. The condition also shows why a simple deposit-beta forecast is insufficient: the pricing decision needs the dollar response, the margin on those dollars, and the stability of those dollars.

6.3 Integrating Pricing with Discrete Choice

In the Section 5 discrete-choice model, separate cost-like price terms from quality and safety attributes. Let the effective deposit-price vector be

$$C_{ijmt}(o) = (r_{ijmt}(o), f_{ijmt}(o), \text{minbal}_{ijmt}(o), \text{feecredit}_{ijmt}(o)),$$

where r is the deposit rate or earnings credit, f are fees, minbal is the minimum-balance requirement, and feecredit captures explicit or implicit relationship pricing. Let

$$Q_{jmt}(o) = (\text{safety}_{jmt}(o), \text{service}_{jmt}(o), \text{digital}_{jmt}(o), \text{treasury}_{jmt}(o))$$

collect quality attributes. The safety component includes deposit insurance structure, collateralization, sweep arrangements, operational continuity, perceived bank soundness, and relationship-manager credibility. Classic bank-run theory makes the safety and liquidity dimension central [22]; depositor-level evidence also shows that relationships and depositor networks matter during runs [35]. The utility for deposit relationship j is

$$U_{ijmt}(o) = \delta_{jmt} + \beta'_i A_{jmt} + \zeta'_i Q_{jmt}(o) - \alpha'_i C_{ijmt}(o) + \kappa'_i B_{ijmt} + \xi_{jmt} + \epsilon_{ijmt}.$$

The focal-bank share response is then

$$s_{ift}(o) = \sum_{j \in \mathcal{J}_{fmt}} \Pr(c_{it} = j \mid C_{ijmt}(o), Q_{jmt}(o), A_{jmt}, B_{ijmt}, \mathcal{I}_{it}).$$

The balance response to pricing is not the same as the choice probability. It is the product of total demand and share:

$$\frac{\partial D_{ift}}{\partial r_{io}} = \frac{\partial D_{it}^*}{\partial r_{io}} s_{ift} + D_{it}^* \frac{\partial s_{ift}}{\partial r_{io}}.$$

For ordinary operating balances, the first term may be small if the price mainly reallocates balances across banks. During stress, the first term can be material if the pricing or safety signal affects the SME's desired liquidity. This equation is the reason stress pricing must be combined with the main dynamic-demand and discrete-choice model.

The same equation also clarifies how to distinguish three common pricing stories. If $\partial D_{it}^* / \partial r_{io} \approx 0$ and $\partial s_{ift} / \partial r_{io} > 0$, pricing mainly reallocates existing SME liquidity from competitors. If both terms are positive, the offer both attracts share and induces larger desired balances, perhaps through perceived liquidity support or safety. If the share term is positive but the post-offer survival of balances is low, the bank has bought rate-sensitive hot money rather than a durable operating relationship.

6.4 Heterogeneous Rate Sensitivity and Deposit Beta

A reduced-form pricing equation useful for implementation is

$$\log D_{ifmt} = \alpha_i + \delta_m + \lambda_t + \beta_{r,i}(R_t) r_{ifmt} + \beta_c C_{fmt} + \beta_q Q_{fmt} + \beta_b B_{ift} + u_{ifmt},$$

where m is a local market, R_t is the regime, C_{fmt} is local competition, Q_{fmt} captures branch, digital, and service quality, and B_{ift} captures relationship depth and switching frictions. The rate coefficient should vary by customer and regime:

$$\beta_{r,i}(R_t) = \beta_0 + \beta'_x X_i + \beta'_m M_m + \beta_R R_t + \beta'_{xR}(X_i \otimes R_t) + \zeta_i.$$

This specification lets the model represent SMEs whose operating balances are rate insensitive in normal periods but become rate sensitive during liquidity competition, as well as SMEs that prioritize safety or operating continuity over rate. Deposit-competition and monetary-policy papers motivate this focus on market power, rate pass-through, and deposit betas [24, 26, 32, 50]; structural banking demand papers motivate the explicit role of prices, attributes, and local competition [3, 23].

For implementation, it is important to separate the *deposit beta* from the *customer balance elasticity*. A deposit beta is a pass-through object, for example

$$b_{if}^r = \frac{\partial r_{ifmt}}{\partial r_t^{\text{mkt}}} \quad \text{or} \quad \frac{\Delta r_{ifmt}}{\Delta r_t^{\text{mkt}}},$$

where r_t^{mkt} is a market policy or wholesale-rate benchmark. The balance elasticity is

$$\varepsilon_{if}^r = \frac{\partial \log D_{ifmt}^r}{\partial r_{ifmt}}.$$

Stress pricing needs both. A high beta can be a defensive pass-through policy with little balance response, while a high balance elasticity can imply a large response to small pricing differences. The recommended model should therefore estimate a response surface

$$\varepsilon_{if}^r = e(X_i, M_m, R_t, \text{Safety}_{f,m,t}, \text{SwitchCost}_{if,t}, \text{Uninsured}_{i,t}),$$

where $\text{Uninsured}_{i,t}$ measures the portion of balances exposed to deposit-insurance limits or uninsured operational risk when the bank has such data or can approximate it. Monetary-tightening stress and uninsured depositor run evidence make this exposure especially relevant [36].

6.5 Branch Market Share as a Noisy Anchor

The bank may observe branch-level deposit market share at granular geography, but such shares usually mix consumer, SME, commercial, and wholesale deposits. They should therefore anchor local competition and broad share, not be treated as direct SME labels. Let $\mathcal{G}$ denote segments such as consumer, SME, commercial, and wholesale. A measurement equation is

$$S_{fmt}^{\text{branch}} = \sum_{g \in \mathcal{G}} \omega_{gmt} S_{fgmt} + e_{fmt},$$

where S_{fgmt} is segment-specific focal-bank share and ω_{gmt} is the local deposit mix. If ω_{gmt} is not directly observed, the model should estimate it using branch location, account mix, relationship-manager portfolio, public market data, and internal segment tags. The deployment output should distinguish:

$$\Pr(S_{f,\text{SME},m,t} \mid S_{fmt}^{\text{branch}}, \mathcal{I}_{mt}) \quad \text{from} \quad S_{fmt}^{\text{branch}}.$$

Confusing these two objects would overstate how much branch-level share identifies SME wallet share.

6.6 Stress Regimes and Deposit Demand

Stress conditions change both total deposit demand and focal-bank share. Let

$$R_t \in \{\text{normal, rate shock, credit stress, liquidity stress, flight to safety}\}$$

denote the regime. The total-demand layer can be written as

$$\begin{aligned} \log D_{i,t+h}^* &= \rho \log D_{it}^* + \gamma' X_{i,t+h} + \lambda_{g(i),t+h} + \theta'_R R_{t+h} \\ &\quad + \theta'_E \text{Exposure}_{i,t+h} + \theta'_L \text{Liquidity}_{i,t+h} + \eta_i + \nu_{i,t+h}. \end{aligned}$$

The share layer should then allow the same regime to alter price sensitivity, safety preference, and switching frictions. It is cleaner to write the stress utility directly from primitives rather than adding stress terms to a baseline utility that may already contain price:

$$\begin{aligned} U_{ijm,t+h}^R(o) &= \delta_{jm,t+h}^R + \beta_i^{R'} A_{jm,t+h} + \zeta_i^{R'} Q_{jm,t+h}(o) - \alpha_i^{R'} C_{ijm,t+h}(o) \\ &\quad + \kappa_i^{R'} B_{ijm,t+h} + \chi'_C(R_{t+h} \otimes C_{ijm,t+h}(o)) \\ &\quad + \chi'_Q(R_{t+h} \otimes Q_{jm,t+h}(o)) + \chi'_S(R_{t+h} \otimes \text{SwitchCost}_{ijm,t+h}) + \xi_{jm,t+h}^R + \epsilon_{ijm,t+h}. \end{aligned}$$

Any regime term common to all alternatives is normalized into the outside option or omitted, because it cancels in a standard logit share. A regime matters for share only through alternative-specific attributes, interactions, or an explicit outside-option normalization. During stress, the sign of deposit change is not universal. Some SMEs may build balances through precautionary saving, policy inflows, or credit-line draws; others may burn cash as revenue falls. Evidence from COVID-period corporate studies documents liquidity draws and a “dash for cash” [2, 43]; broader theory links bank lending commitments and deposit-taking [39]; and credit-line research treats lines as liquidity insurance [1, 59]. For SMEs, the model should estimate this heterogeneity by industry, cash-flow exposure, credit access, relationship depth, and geography rather than imposing a single stress multiplier.

For stress pricing, the regime variable should not be only a historical label. It should be part of a scenario generator. One implementation is a transition model

$$\Pr(R_{t+1} = k \mid R_t, \mathcal{Z}_t) = \frac{\exp(a_k R_t + b'_k \mathcal{Z}_t)}{\sum_{\ell} \exp(a_{\ell} R_t + b'_{\ell} \mathcal{Z}_t)},$$

where $\mathcal{Z}_t$ includes market rates, local unemployment, industry stress, credit spreads, deposit outflow indicators, competitor rate gaps, and bank-specific safety indicators. A management stress scenario can override this transition model by conditioning on a path $\{R_{t+1}, \dots, R_{t+H}\}$ and corresponding paths for rates, credit conditions, and competitor actions. This keeps the statistical model and the management scenario internally consistent.

6.7 Runoff, Duration, and Stable Balance Value

Stress-condition pricing is incomplete without a runoff model. Let T_{io}^r be the time until the incremental balance generated by offer o runs off in scenario r . Define the survival curve

$$S_{io}^r(h) = \Pr(T_{io}^r > h \mid \mathcal{I}_{it}, o, R_{t:t+h} = r)$$

and the stable-balance contribution

$$\text{SBD}_{io}^r(H) = \sum_{h=1}^H \delta^h S_{io}^r(h) \Delta D_{ifo,0}^{\text{move},r}.$$

Here $\Delta D_{ifo,0}^{\text{move},r}$ is the originated moved balance before runoff. If the model instead forecasts a realized post-runoff path $\Delta D_{ifo,t+h}^{\text{move},r}$, then the stable-balance contribution should be $\sum_h \delta^h \Delta D_{ifo,t+h}^{\text{move},r}$.

without another multiplication by $S_{io}^r(h)$. Mixing these two conventions double counts runoff. The model should estimate separate survival curves for moved competitor balances, expanded liquidity demand, and cannibalized balances:

$$S_{io}^{k,r}(h), \quad k \in \{\text{move, expand, cann}\}.$$

This distinction matters because a rate promotion may generate large first-week or first-month inflows but weak stable-balance value. A practical hazard model is

$$\begin{aligned} \Pr(T_{io}^{k,r} = h \mid T_{io}^{k,r} \geq h, \mathcal{I}_{it}) \\ = \Lambda(a_{krh} + b'_{kr}X_i + c'_{kr}\text{Offer}_{io} + d'_{kr}\text{CompetitorGap}_{mt+h} + e'_{kr}\text{Relationship}_{if,t+h}), \end{aligned}$$

where Λ is a logit or complementary-log-log link. The pricing objective should value balances by expected survival, not merely by day-one inflow. Thus a serious stress-pricing report should show both $\widehat{\Delta D}_{ifo}^r(H)$ and $\widehat{\text{SBD}}_{io}^r(H)$.

6.8 Stress Offer Design: Rate, Safety, and Relationship Terms

In stress conditions, the bank should not treat “price” as only the posted rate. The action vector should be

$$o = (r, \text{fee, earnings credit, sweep, collateral, operating support, relationship terms}).$$

For some SMEs, especially operating-account customers with payroll, payables, and merchant settlement needs, reliability, funds availability, credit-line access, and deposit-insurance or sweep structure can dominate a small rate difference. For other SMEs with excess cash and low switching costs, the rate gap may dominate. The model should therefore estimate substitution between rate and safety or relationship terms, not only the level effect of rate. A useful willingness-to-pay contrast is

$$\text{WTP}_i^{\text{safety},r} = \frac{\partial U_{ijmt}^R / \partial Q_{jmt}^{\text{safety}}}{\partial U_{ijmt}^R / \partial r_{ijmt}},$$

interpreted as the rate concession or premium that holds utility fixed for a one-unit safety improvement, with the sign determined by the rate convention. If higher r is better for depositors, the denominator is positive; if r is entered as a bank cost, the sign should be reversed. The scale of the safety measure must be reported for the WTP number to be meaningful. This connects the stress extension back to the discrete-choice model and to Ma’s individual-specific financial-product pricing framework [46].

6.9 Scenario Outputs for Management

For each stress scenario r and offer o , the model should produce a decomposition:

$$\begin{aligned} \widehat{\Delta D}_{ifo}^r(H) &= \widehat{D}_{i,t+H}^{*,r}(0) \widehat{\Delta s}_{ifo}^r(H) + \widehat{s}_{if,t+H}^r(0) \widehat{\Delta D}_{io}^{*,r}(H) \\ &\quad + \widehat{\Delta s}_{ifo}^r(H) \widehat{\Delta D}_{io}^{*,r}(H), \end{aligned}$$

where the first term is share capture, the second is total-demand expansion, and the third is interaction. Management should see these as separate numbers, not only a single incremental-balance forecast. The pricing package should also report:

$$\{\widehat{\Delta D}^r, \widehat{\text{SBD}}^r, \widehat{\text{NPV}}^r, \widehat{\text{Runoff}}^r, \widehat{\text{Cannibalization}}^r, \widehat{\text{LiquidityValue}}^r, \widehat{\text{ConfidenceInterval}}^r\}.$$

This output answers the practical question: whether a higher offered rate is buying durable SME operating deposits, temporary hot money, balances that would have arrived without pricing, or balances cannibalized from existing focal-bank relationships.

The scenario package should also report break-even pricing. Because demand, runoff, cannibalization, and relationship spillovers can make value nonmonotone in the offered rate, the primary object should be the acceptable-rate set:

$$\mathcal{R}_{io}^{\text{BE},r} = \{r \in \mathcal{R}_{io} : \hat{J}_{io}^{\text{price},r}(H) \geq 0\}.$$

Only when this set is an interval, or when the model explicitly assumes local monotonicity over the feasible offer range, should the report summarize it by $r_{io}^{\text{BE},r} = \sup \mathcal{R}_{io}^{\text{BE},r}$. If the proposed offer rate lies outside $\mathcal{R}_{io}^{\text{BE},r}$, the model should explain which assumption makes the action still acceptable, such as relationship profit, strategic retention, liquidity value, or a management override. If no such assumption is documented, the offer is not economically justified even if it is expected to grow balances.

6.10 Identification and Validation Requirements

Stress pricing requires stronger evidence than normal-period ranking. The key identification threats are endogenous pricing, endogenous relationship-manager attention, unobserved competitor rates, segment mixing in branch market share, and regime-specific selection. A serious implementation should use at least one of the following sources of identifying variation:

1. randomized or quasi-random deposit-rate and fee-waiver tests;
2. rate-sheet changes or policy thresholds that create exogenous price variation;
3. competitor-rate shocks or local market shocks used as structural demand shifters, or as instruments only when an explicit exclusion restriction is credible after conditioning on local demand and quality;
4. branch or relationship-manager capacity variation that affects contact but not latent deposit demand after controls;
5. pre/post conversion evidence that identifies gross migrated balances, post-stress runoff, and cannibalization.

Validation should include normal and stress windows separately. Required gates are:

$$\text{Bias}_{gr} = \frac{1}{|V_{gr}|} \sum_{i \in V_{gr}} \frac{\hat{D}_i^r - D_i^r}{D_i^r}, \quad \text{Coverage}_{gr}(\alpha) = \frac{1}{|V_{gr}|} \sum_{i \in V_{gr}} 1\{D_i^r \in I_i^{r,1-\alpha}\},$$

for segment g and regime r . The pricing model should not be used for material stress pricing unless dollar calibration, interval coverage, rate-elasticity signs, runoff forecasts, and cannibalization decomposition pass in the relevant segment. If those gates fail, the model can still support learning-cell design and management overlays, but not automated stress pricing.

For governance, stress pricing should have explicit promotion thresholds:

$$\begin{aligned} |\text{Bias}_{gr}| &\leq b_{\max}, & |\text{Coverage}_{gr}(\alpha) - (1 - \alpha)| &\leq c_{\max}, \\ \text{sign}(\hat{\varepsilon}_{if}^r) &\text{ stable across accepted specifications,} & \Pr(\widehat{\text{NPV}}_{io}^r < 0) &\leq q_{\max}, \\ \left| \widehat{\text{Runoff}}_{gr}^{\text{stress}} - \text{Runoff}_{gr}^{\text{realized}} \right| &\leq u_{\max}. \end{aligned}$$

The last line is especially important: a model that captures first-month inflows but misses post-stress decay can overstate the value of stress pricing. When historical stress observations are sparse, validation should use backtests across rate-hiking cycles, local market shocks, industry downturns, competitor rate changes, and deliberately randomized learning cells. The result should be labeled by evidence grade: randomized, quasi-experimental, historically backtested, scenario-only, or management overlay.

7 Estimation Problems Specific to the Recommended Model

The proposed architecture is useful only if the bank estimates the right objects. A large premium bank faces a particularly important identification problem: its observed SME customers are not a random sample of the SME market. They may be larger, more complex, more urban, more digitally active, more relationship-oriented, more multi-product, or more sensitive to brand and operational reliability than SMEs served by regional and community banks. If the model is fit only on focal-bank customers, it can learn the behavior of SMEs that already selected a premium bank, not the behavior of competitor-bank customers the bank wants to win.

This section states the estimation problems in the notation of Sections 5 and 6. The main point is that selection, unobserved quality, and endogenous prices enter every component of

$$D_{ift}(o, M_t, R_t) = D_{it}^*(M_t, R_t) s_{ift}(o, M_t, R_t),$$

and therefore directly affect estimated competitor wallet, pricing response, move-over, and risk-adjusted value.

The closest literatures each solve part of the problem, but none solves the full SME wallet-acquisition problem alone. BLP-style banking demand models use market shares, product attributes, and instruments to handle unobserved product-market quality [9, 23, 51]. Ma extends this logic to financial products with individual-specific prices and selectively observed chosen-product prices [46]. Sample selection and external-validity literatures clarify why estimates from one selected population need not transport to another [33, 34, 47]. Causal response methods separate propensity from incremental treatment effects [17, 58, 66]. Our setting must combine all of these: a selected premium-bank panel, latent competitor wallets, negotiated SME prices, branch-level aggregate anchors, and campaign interventions.

7.1 Premium-Bank Selection and Transportability

Let $B_i = f$ denote that SME i is observed as a focal-bank customer. The bank usually estimates on

$$p(Y_i, X_i \mid B_i = f),$$

but deployment requires predictions for prospects currently held by competitors:

$$p(Y_i(o), X_i \mid B_i \neq f).$$

The selection problem is that

$$p(h_{it} \mid X_i, B_i = f) \neq p(h_{it} \mid X_i, B_i \neq f),$$

where h_{it} is the latent state containing liquidity need, price sensitivity, switching cost, service preference, and relationship depth. A Chase-like bank's customers may have systematically higher

Model component	What the literature typically does	Why it is insufficient here	Required adaptation
Share model s_{ift}	BLP/micro-BLP uses market shares, product attributes, and instruments for prices	SME bank quality and relationship value are partly unobserved and premium-bank clients are selected	Product-market quality controls, instruments, and target-population diagnostics
Price vector C_{ijmt}	Financial-product demand models allow individual-specific prices and selected observed prices	SME prices include negotiated rates, fee waivers, treasury terms, and rejected quotes	Quote ledger, price-imputation validation, and selectivity correction
Wallet W_{ipt}^*	Cross-sell and latent-factor models infer missing product ownership or amount	Competitor dollar balances are latent and branch shares mix segments	External anchors, segment-mix measurement equations, and uncertainty intervals
Dynamic demand D_{it}^*	Dynamic panels forecast balances from lags and market conditions	Premium-bank customers may have different growth, seasonality, and stress behavior	Stratified out-of-time tests and separation of demand growth from share movement
Move-over and policy J_{io}	Uplift and causal models estimate treatment effects under assignment assumptions	RM attention, pricing exceptions, and campaigns are targeted	Randomized learning cells, overlap diagnostics, and orthogonal scores

Table 2: How existing estimation approaches map into the recommended SME wallet model.

operating complexity and stronger preference for scale, branch/digital reliability, treasury integration, and brand safety. A model trained only on these customers may overstate the competitor-bank prospect’s willingness to move for the same offer, or understate the rate premium needed to dislodge a relationship at a regional bank.

The remedy is not to “adjust for bank size” with one control variable. The model needs a target-population correction. Let $T_i = 1$ denote the target population for a campaign or prospect universe, and let $S_i = 1$ denote membership in the estimation sample. A transparent first correction is importance weighting:

$$w_i = \frac{\Pr(T_i = 1 \mid X_i)}{\Pr(S_i = 1 \mid X_i)}.$$

The weighted estimating equations for any module parameter θ become

$$\sum_{i,t} w_i \psi(Y_{it}, X_{it}; \theta) = 0.$$

This is the analogue of reweighting or transporting estimates from one sample to another in program-evaluation settings [34]. It helps only for selection on observed variables. If unobserved service preference or switching cost differs between premium-bank customers and regional-bank customers, the problem becomes partially identified; the honest output is a sensitivity range or bound, not a single point estimate [47]. Therefore the bank should also use external anchor samples: consented account aggregation, onboarding statements, win/loss files, branch-market aggregates, market-cell benchmarks, and conversion events. The target-population diagnostic is segment calibration:

$$\text{Bias}_{gp}^T = \frac{1}{|V_{gp}^T|} \sum_{i \in V_{gp}^T} \frac{\widehat{W}_{ipt}^* - W_{ipt}^A}{W_{ipt}^A},$$

computed on validation samples that resemble the prospect population, not only current focal-bank customers. If this diagnostic fails, the model should be reported as a within-bank relationship-deepening model, not as a competitor wallet-acquisition model.

7.2 Unobserved Bank Quality in the Share Model

The discrete-choice layer contains unobserved bank-market quality:

$$U_{ijmt} = \delta_{jmt} + \beta'_i A_{jmt} + \zeta'_i Q_{jmt} - \alpha'_i C_{ijmt} + \kappa'_i B_{ijmt} + \xi_{jmt} + \epsilon_{ijmt}.$$

For a premium bank, ξ_{jmt} is not a nuisance term. It may include brand trust, perceived safety, treasury-service reliability, digital uptime, relationship-manager quality, geographic convenience for owners and employees, and the value of being with a systemically important bank. These attributes are partly observed by customers and relationship managers but not fully observed by the econometrician. If cost-like prices or offers are correlated with ξ_{jmt} , then

$$\mathbb{E}[C_{ijmt}\xi_{jmt}] \neq 0,$$

and the estimated price coefficient confounds price sensitivity with unobserved quality. This is the standard differentiated-product demand issue in BLP-style models [9, 51], and it is especially important in banking because service and safety quality are central parts of the product.

The model should therefore estimate product-market fixed effects or mean utilities δ_{jmt} , use instruments or control functions for endogenous price, and include explicit quality proxies where possible. Candidate quality controls include branch density, digital-service incidents, service response times, relationship-manager coverage, treasury implementation time, complaint rates, local deposit outflow stress, bank credit ratings, uninsured-deposit exposure, and customer-service capacity. Candidate instruments include rate-sheet rules, funding-cost shifters, policy thresholds, centralized pricing changes, and cost interactions with product attributes. Competitor-rate shocks should usually enter as demand-environment variables or competitor-response scenarios; they are instruments only with a documented exclusion restriction. The identifying moment is

$$\mathbb{E}[Z_{jmt}\xi_{jmt}(\theta)] = 0.$$

This is the same role played by instruments in differentiated-product demand estimation: they separate price variation caused by supply or cost shifters from price variation caused by unobserved demand quality [9, 51]. In banking applications such as Dick [23] and Egan et al. [26], local competition, product attributes, and banking-market structure provide part of that identifying variation. For SME wallet capture, the additional difficulty is that the premium-bank brand and relationship quality are themselves central attributes, so the quality controls must be richer than in a simple posted-rate deposit model. When that moment is not credible, the output should be labeled as descriptive preference or association. It should not be used for rate optimization. Control-function approaches are another way to handle endogenous prices in choice models [55, 67].

7.3 Individual-Specific Prices and Selectively Observed Quotes

For SME banking, the effective cost-like price vector

$$C_{ijmt}(o) = (r, f, \text{minbal}, \text{feecredit})$$

is often negotiated; safety and service belong in the quality vector $Q_{jmt}(o)$. The econometrician may observe booked prices for chosen products and perhaps some rejected quotes, but not the

full menu of prices each SME would have faced at all competitors. This is exactly why the Ma framework is relevant: in financial products, customers can face individual-specific prices, and using a common market price can bias estimated price sensitivity when those prices are correlated with demographics, risk, or income [46].

For the recommended model, the first-stage pricing equation should be written for every feasible bank-product alternative:

$$C_{ijmt}^k = q_k(Z_{ijmt}^P, X_i, A_{jmt}, M_t) + v_{ijmt}^k, \quad k \in \{r, f, \text{minbal}, \text{feecredit}\}.$$

The problem is that C_{ijmt}^k is often observed only when a quote is generated or a product is chosen. Let $R_{ijmt}^P = 1$ indicate observed price. Then

$$R_{ijmt}^P = 1\{\pi' Z_{ijmt}^R + \nu_{ijmt} > 0\}.$$

If quote observation depends on latent demand or bargaining ability, $\mathbb{E}[v_{ijmt}^k \mid R_{ijmt}^P = 1] \neq 0$. Ma's estimator treats this as a selectivity problem distinct from the classic Heckman market-entry problem: the customer is in the market, but the econometrician observes only the chosen product's individualized price [33, 46]. In our SME setting, the same issue appears when we observe the booked treasury or deposit terms but not the competitor terms and rejected focal-bank quotes. The remedy is to maintain a quote ledger that records offered, rejected, expired, and exception-approved terms, and to use selectivity correction or control functions when unchosen prices must be imputed. For governance, the price-imputation layer should have its own out-of-sample loss:

$$L_P = \frac{1}{|Q|} \sum_{(i,j,m,t) \in Q} \|C_{ijmt}^{\text{obs}} - \hat{C}_{ijmt}\|^2,$$

where Q is a held-out quote sample. Poor price prediction should veto counterfactual pricing even if the final choice model has good in-sample fit.

7.4 Latent Competitor Wallet and Segment Mixing

The model uses latent competitor wallet

$$A_{ipt}^{-f} = W_{ipt}^* - a_{ipt}^{\text{obs}}.$$

This object is not directly observed for most prospects. Public and branch market-share data help anchor local totals, but they mix consumer, SME, commercial, and wholesale balances. Thus the measurement equation is

$$S_{fmt}^{\text{branch}} = \sum_{g \in \mathcal{G}} \omega_{gmt} S_{fgmt} + e_{fmt},$$

not $S_{f,\text{SME},m,t} = S_{fmt}^{\text{branch}}$. If the segment weights ω_{gmt} are misspecified, the model can incorrectly infer that a branch market is weak in SME share when the weakness is actually consumer deposits, or vice versa.

The remedy is a hierarchical measurement model with sensitivity analysis over segment-mix assumptions:

$$p(W^*, S, \omega \mid a^{\text{obs}}, S^{\text{branch}}, A^{\text{anchor}}, X).$$

This is closest in spirit to latent-trait and mixed-data factor models in the cross-selling literature [37, 38], but the SME wallet version has a harder level-identification problem because the missing

variable is a dollar competitor balance, not only an unobserved product affinity. EM or Bayesian data augmentation can estimate the latent structure [21], but the level of W_{ipt}^* still requires anchors. The output should include posterior or bootstrap intervals for A_{ipt}^{-f} , not only a point estimate. Management reports should distinguish three evidence grades: directly anchored competitor wallet, market-anchored competitor wallet, and model-imputed competitor wallet. Only the first two should be used for dollar claims in high-stakes targeting.

7.5 Dynamic Demand Versus Share Movement

Observed deposit growth after an offer is

$$\Delta D_{if,t:t+H}^{\text{obs}} = \Delta D_i^{*,H} s_{if}^0 + D_i^{*,0} \Delta s_{if}^H + \Delta D_i^{*,H} \Delta s_{if}^H + u_{iH}.$$

The first term is growth in total liquidity demand, the second is share movement, and the third is interaction. In a premium bank, large clients may grow faster or experience different seasonality than regional-bank clients. If the model attributes all observed balance growth to Δs_{if}^H , it will overstate competitor capture. If it attributes all stress-period balance growth to D_i^* , it will understate the effect of safety and relationship share.

The remedy is to estimate the dynamic demand layer and share layer jointly, or to use a two-step estimator with generated-regressor uncertainty carried forward. Dynamic-panel methods such as Arellano and Bond [5] address persistence and fixed effects, while corporate-liquidity studies motivate why cash and deposits vary with financing constraints and shocks [4, 7, 53]. Those literatures do not identify competitor share movement by themselves. Required diagnostics include forecast error for D_i^* , share-anchor residuals, and reconciliation of observed balance changes into move, expansion, and cannibalization. Out-of-time tests should be stratified by incumbent-bank type, prospect channel, industry, geography, and relationship size.

7.6 Campaign Assignment, Relationship-Manager Attention, and Uplift

The action o is usually not random. Relationship managers contact customers who look promising, price exceptions are granted to strategically important firms, and premium banks may selectively pursue clients already likely to move. Thus

$$O_i \not\perp \{Y_i(o) : o \in \mathcal{O}\} \mid \mathcal{I}_{it}$$

can fail. In that case, propensity, predicted wallet, and treatment effect are not the same object.

The preferred remedy is randomized or quasi-random learning cells for deposit offers, pricing, onboarding support, and consolidation campaigns. When random assignment is not feasible, use overlap diagnostics and orthogonal scores:

$$\begin{aligned} \psi_i(o) = & m_o(X_i) - m_0(X_i) + \frac{1\{O_i = o\}}{e_o(X_i)} \{Y_i - m_o(X_i)\} \\ & - \frac{1\{O_i = 0\}}{e_0(X_i)} \{Y_i - m_0(X_i)\}. \end{aligned}$$

This score is credible only if overlap holds and the conditioning set contains the variables that drove assignment. This is the same distinction made in the uplift and causal-inference literatures between response propensity and incremental response [17, 45, 58, 66]. The model should report the effective sample size after weighting, standardized differences before and after adjustment, placebo effects on pre-treatment balances, and lift in holdout cells. Otherwise, the model may be used for prioritization but not causal incrementality.

7.7 Equilibrium and Competitor Response

Counterfactual pricing assumes something about competitor behavior. If the focal bank raises rates for selected SMEs, competitors may match, selectively retain high-value accounts, or respond through service and credit terms. The share simulation

$$s_{ift}(o)$$

is therefore conditional on a competitor-response scenario. A unilateral simulation is useful for small tests but can overstate capture in broad campaigns.

The remedy is to define counterfactual regimes explicitly:

$$s_{ift}^r(o, o_{-f}^0), \quad s_{ift}^r(o, o_{-f}^{\text{match}}), \quad s_{ift}^r(o, o_{-f}^{\text{retain}}).$$

Management should see wallet capture under no-response, rate-match, and selective-retention scenarios. A recommendation that works only under no-response assumptions should be treated as a testable hypothesis, not a base case.

7.8 Minimum Estimation Standard

Before the recommended model is used for material targeting or pricing, the bank should require the following estimation ledger:

1. **Selection ledger:** comparison of current focal-bank customers, target prospects, won prospects, and lost prospects; target-population weights and sensitivity to unobserved selection.
2. **Price ledger:** quote coverage, rejected quotes, exception rates, price-imputation accuracy, and evidence on price exogeneity or instruments.
3. **Quality ledger:** observed quality proxies, unobserved product-market fixed effects, and sensitivity of price elasticities to quality controls.
4. **Wallet ledger:** source of each competitor-wallet estimate: directly observed, anchored, or imputed.
5. **Dynamic ledger:** decomposition of balance changes into total liquidity demand, share movement, interaction, cannibalization, and residual.
6. **Causal ledger:** randomized evidence, quasi-random evidence, or associational status for each offer and segment.
7. **Counterfactual ledger:** competitor-response scenario, stress regime, price assumptions, and uncertainty interval for every policy output.

These ledgers make the estimation problem auditable. They also prevent the largest practical mistake in this project: presenting a precise-looking wallet or pricing number when the identifying evidence supports only a ranking, a sensitivity range, or a learning-cell recommendation.

8 Validation and Evaluation

Validation should be separated into ledgers that match the estimands.

Wallet reconstruction. Evaluate whether estimated wallet sizes match external validation samples, customer self-reports, bureau data, account aggregation data, or post-acquisition observations when a customer consolidates accounts. Metrics include calibration, rank ordering, interval coverage, and error by industry and firm size.

Product amount and severity. Evaluate the conditional amount model separately from product incidence. Metrics should include balance-weighted error, mean and quantile calibration, prediction interval coverage, segment-level relative bias, and retransformation diagnostics for log-dollar models. A high adoption AUC does not validate estimated deposit balances, loan balances, card spend, or assets under management.

Dynamic deposit demand. Evaluate deposit-balance forecasts out of time and across stress regimes. Metrics should include calibration of prediction intervals, error by industry and season, stress-period stability, and decomposition of forecast error into total-liquidity demand error versus focal-bank share error. A model that ranks large deposit customers well in normal times may still fail if it treats stress liquidity as a fixed wallet stock.

Stress-condition pricing. Evaluate rate sensitivity, deposit beta, and runoff separately by stress regime. Metrics should include dollar calibration by scenario, rate-elasticity sign and magnitude stability, branch-share anchor residuals, balance decay after stress, and NPV sensitivity to funding value and pricing cost. Stress pricing should not be promoted from normal-period elasticity alone.

Product prediction. Evaluate product adoption, balance growth, and sequence prediction using out-of-time samples. Metrics include AUC, precision at top k , calibration, balance-weighted error, and next-product hit rate. These are predictive metrics, not proof of causal campaign effectiveness.

Causal response. Evaluate campaigns with randomized holdouts or credible quasi-experiments. Metrics include incremental adoption, incremental balances, risk-adjusted profit, retention, and adverse outcomes. A campaign should not be promoted based only on high propensity scores or backtested correlation.

Move-over decomposition. Evaluate whether predicted moved amounts match observed evidence about migration origin in conversion samples. Metrics include cap violations ($\widehat{\Delta a}^{\text{move}} > \widehat{A}^{-f}$), reconciliation error in the move, expansion, and cannibalization identity; gross and net balance lift; post-offer decay of migrated balances; and product-level diversion from existing focal-bank products. Without this validation, the model should report incremental net balances rather than competitor dollars captured.

9 Research Gaps

Several gaps remain for the exact US SME banking use case.

1. Most direct cross-selling papers use consumer or general financial services data, not modern US SME banking panels.
2. Competitor product holdings and balances are usually unobserved, so validation is difficult.

3. Even when competitor-held balances are estimated, the movable fraction of those balances is usually not identified without conversion, transfer, account-aggregation, or experimental evidence.
4. Deposit size is not merely another product label; it is continuous, rate-sensitive, liquidity-sensitive, and operationally tied to payments. It should be treated as dynamic demand, not as a constant wallet stock.
5. The literature on SME-specific deposit dynamics during stress is less direct than the broader corporate cash, credit-line, and bank-liquidity literature. A bank should validate stress behavior with its own panel data.
6. Marketing propensity and causal treatment effect are frequently conflated in practice.
7. Relationship lending creates joint value across deposits, payments, lending, and monitoring; product-by-product models may understate this complementarity.
8. Product-assortment, next-best-product, and bundle design are useful extensions, but they are not the main problem addressed here. They should be treated as further research after the deposit-wallet and deposit-capture estimands are validated.
9. Compliance, fair-lending, privacy, and model-risk constraints are central in bank deployment but are not the main subject of the academic cross-sell literature.

10 Recommended Implementation Roadmap

A bank can proceed in stages.

Stage 1: Define the wallet taxonomy. Start with a deposit-centered taxonomy: operating deposits, savings and liquidity products, treasury-linked balances, and externally held competitor balances. Define incidence, size, competitor-held amount, movable amount, net incremental amount, runoff, and profitability measures. Other products can be added as relationship context or later extensions.

Stage 2: Build baseline wallet and liquidity proxies. Use internal balances, transaction flows, industry benchmarks, revenue estimates, and market-level deposit data to create transparent baseline wallet estimates. For deposits, construct dynamic liquidity proxies such as trailing cash inflow volatility, payroll coverage months, working-capital intensity, seasonality, and local market position. These baselines are necessary comparators for later machine-learning models.

Stage 3: Fit deposit incidence and size models. Use two-part models for deposit relationships: an incidence model for whether the SME has relevant deposits and a severity model for positive balances. Use other products as covariates or latent relationship context. Report uncertainty intervals, conditional-amount calibration, quantile calibration, and segment-level dollar bias.

Stage 4: Estimate dynamic deposit demand. Fit panel time-series models for deposit balances and total liquidity demand. Include lagged deposits, transaction flows, industry-by-time effects, geography-by-time effects, rate environment, local competition, and relationship terms. Benchmark against seasonal naive and fixed-effects baselines before using more complex sequence models.

Stage 5: Estimate deposit pricing and share response. Use rate, fee, service, branch-market, and relationship variation to estimate deposit share response. Treat branch-level market share as a noisy aggregate anchor when it cannot separate consumer, SME, commercial, and whole-sale deposits.

Stage 6: Stress-test deposit demand. Estimate stress scenarios for balance growth, cash burn, flight-to-safety migration, and post-stress runoff. Validate these scenarios with the bank's own panel history and management judgment.

Stage 7: Introduce randomized campaign holdouts. For each major deposit action, maintain holdouts and estimate incremental lift. Avoid using raw purchase propensity as the promotion criterion.

Stage 8: Estimate move-over and cannibalization. Use conversion events, account-transfer evidence, payoff and balance-transfer records, account aggregation, direct-deposit or ACH migration, and randomized consolidation offers to estimate the fraction of competitor-held balances that move under each offer. Separate migrated competitor wallet from total-demand expansion and cannibalization of existing focal-bank balances.

Stage 9: Optimize for risk-adjusted relationship value. Combine expected incremental balances, fee revenue, interest margin, funding value, credit risk, servicing cost, liquidity value, and compliance constraints.

Stage 10: Monitor drift and fairness. Monitor calibration, response, customer complaints, adverse action patterns, industry/geographic stability, stress-regime performance, and relationship-manager override behavior.

11 Conclusion

There is substantial literature relevant to SME deposit wallet-size estimation, but it is distributed across marketing science, banking economics, SME finance, and causal inference. The direct cross-selling literature provides models for latent product need and wallet context, while share-of-wallet research clarifies the distinction between customer retention and relationship depth. Deposit-demand and bank-competition research explain why deposits require special treatment. Corporate liquidity and credit-line research shows why deposits should not be treated as a fixed-size wallet bucket: firms adjust cash balances and liquidity demand as conditions change. Relationship-lending papers show why SME transaction accounts and credit products are jointly valuable. Causal uplift methods are needed to decide which deposit actions will actually move wallet.

The recommended solution is therefore not a single score. It is a modular decision system: latent wallet estimation, dynamic deposit demand, focal-bank share response, deposit pricing, campaign uplift, and constrained profit optimization. Such a system should be validated with external or

post-conversion wallet evidence, branch-market-share anchors used cautiously, out-of-time deposit forecasts, stress-regime diagnostics, and randomized marketing holdouts before it is used for major pricing or targeting decisions.

Scope Note

This document is a technical literature synthesis for problem formulation and model design, not a full systematic review. It emphasizes sources that are directly useful for banking cross-selling, wallet share, deposit competition, SME relationship lending, dynamic liquidity demand, and causal response. Before using the document as a formal literature-review artifact, the bank should complete a documented source-support ledger, backward and forward snowballing, retraction/erratum checks, and an omission-risk register.

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